



Available online at www.ijournalse.org

Emerging Science Journal

(ISSN: 2610-9182)

Vol. x, No. x, xxxxxxxxxx, 20xx



Fault Signature Maps: A Signal-Level Explainable Artificial Intelligence Framework for Bearing Fault Diagnosis

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Abstract

Deep learning models for bearing fault diagnosis on the Case Western Reserve University (CWRU) dataset routinely achieve near-perfect accuracy. Yet their decisions remain largely opaque at the signal level, so engineers cannot determine where in the raw vibration waveform the model focuses. This study aims to bridge the interpretability gap in deep learning-based bearing fault diagnosis by developing a signal-level explainability framework for the Wide Deep Convolutional Neural Network (WDCNN) on the Case Western Reserve University (CWRU) dataset. SHAP DeepExplainer is applied directly to raw 2,048-point vibration segments to produce per-sample-point attribution maps; Fault Signature Maps (FSMs) are formalized as class-averaged SHAP fingerprints in three variants (signed, absolute, variance) and validated via discriminability index, severity monotonicity, and split-half stability, complemented by a three-variant ablation study examining how architectural decisions affect both accuracy and explainability. On the 10-class CWRU dataset under a rigorous temporal split (1,240/310/750 samples), WDCNN achieves 99.87% test accuracy with a macro F1-score of 0.997. FSMs demonstrate high reproducibility (split-half stability = 0.940), severity monotonicity ranges from 8.6% to 17.6%, and ablation reveals that removing Batch Normalization increases FSM discriminability by 2.3× at a 3.6% accuracy cost. As the first work applying SHAP DeepExplainer at full 2,048-point raw signal resolution for WDCNN, this study establishes that fault discrimination relies on transient impulse morphology rather than bearing characteristic frequencies, a finding invisible to feature-level XAI, and introduces a multi-resolution diagnostic framework bridging deep learning accuracy with physically interpretable vibration analysis for trustworthy deployment in safety-critical industrial environments.

Keywords: Bearing Fault Diagnosis; WDCNN; SHAP; Explainable AI; Signal-Level Attribution; CWRU Dataset; Predictive Maintenance; Deep Learning Interpretability; Fault Signature Maps.

Article History:

Received:

Revised:

Accepted:

Published:

1- Introduction

Rolling-element bearings are among the most mechanically critical and failure-prone components in modern industrial rotating machinery, with bearing degradation accounting for 40-51% of all rotating equipment failures across manufacturing, energy, aerospace, and transportation sectors [1][2][3]. Unplanned bearing failures trigger cascading consequences, such as catastrophic equipment damage, production-line stoppages, escalating maintenance costs, and serious worker safety hazards that extend well beyond simple component replacement costs to encompass lost production capacity, emergency procurement, and potential contractual penalties. Traditional maintenance strategies, reactive (run-to-failure) and fixed-interval preventive, are demonstrably inadequate for these demands [3]. Reactive maintenance

exposes facilities to unpredictable stoppages; fixed-interval maintenance incurs unnecessary replacement costs without eliminating unexpected in-cycle failures. Predictive maintenance (PdM), anchored in continuous vibration-based condition monitoring and data-driven fault classification, offers a compelling alternative: maintenance actions aligned with actual component health rather than arbitrary schedules [1][3].

The Case Western Reserve University (CWRU) bearing dataset [4] has served for over two decades as the field's canonical benchmark, providing controlled laboratory vibration signals from seeded inner-race, outer-race, and rolling-element faults at three severity levels and multiple load conditions. This standardized benchmark has driven successive waves of algorithmic innovation, enabling rigorous cross-study comparisons of signal-processing and machine-learning methods. Deep learning has transformed CWRU-based bearing fault diagnosis over the past decade. Hierarchical Adaptive Deep CNN architectures achieve 99.91% accuracy [5]; CNN-LSTM hybrids that model both local spectral patterns and temporal evolution reach 99.6% [6]; ResNet-50 transfer learning from ImageNet spectrograms attains 99.99% [7]. Attention-based mechanisms and comprehensive architectural surveys further confirm the technical maturity of deep learning for this task [24]. More recently, Xu et al. systematically evaluated five distinct CNN designs — SerConv, ResConv, OSAConv, CSAConv, and MD-DAConv — integrating multi-directional and multi-scale convolutions with residual connections to improve robustness on both CWRU and SJTU-SY datasets [30]. To address the persistent challenge of strong noise in realistic industrial environments, Zhang et al. introduced an MSCNN-GRU framework augmented by a convolutional attention module (CBAM), demonstrating superior generalization across varying load and speed conditions [31], complemented by a companion MLSCA-CW network embedding soft-threshold noise filtering at multiple spatial locations [32]. Li et al. further broadened the landscape by combining multi-scale permutation entropy with a multi-channel fusion CNN, enabling simultaneous exploitation of multi-sensor information for stable diagnosis on both rolling bearings and planetary gearboxes [33]. Among these architectures, the Wide Deep Convolutional Neural Network (WDCNN) proposed by Zhang et al. has become one of the most widely adopted 1D-CNN baselines, distinguished by its wide first convolutional kernel (length 64, stride 16) that processes raw vibration signals directly without hand-crafted feature engineering [8].

Despite these accuracy achievements, industrial adoption of AI-driven fault detection remains constrained. The limiting factor is not insufficient accuracy but the absence of interpretability: maintenance technicians cannot determine why a model flags a bearing as faulty, which signal characteristics drive the prediction, or whether the model's reasoning aligns with established vibration physics [9][10]. This opacity generates justified hesitation in operations teams who must act on model recommendations without understanding their basis [9]. Regulatory frameworks governing safety-critical sectors such as aviation, nuclear, and rail transport increasingly mandate explainability for automated decision systems, creating a formal barrier to deploying opaque deep learning models [11]. Recent comprehensive reviews of deep learning for bearing diagnostics confirm that interpretability remains the field's most pressing open challenge, noting that most studies optimize classification metrics while leaving internal representations unexamined [24][29].

1-1- The Accuracy-Interpretability Gap

A fully transparent model achieving 94–96% accuracy that engineers can audit and trust delivers substantially greater industrial value than a 99%-accurate black-box [12]. This tension between accuracy and interpretability motivates the present study. Our previous work addressed this gap in classical machine learning by conducting a comprehensive SHAP (SHapley Additive exPlanations) analysis of optimized Support Vector Classifier (SVC) and Logistic Regression (LR) models trained on nine time-domain statistical features extracted from the CWRU dataset [12]. That study achieved 96.4% and 94.3% accuracy, respectively, and revealed that SVC emphasizes distribution-shape features (crest factor, kurtosis, skewness), whereas LR relies on amplitude-based features (mean, RMS, standard deviation). Only the mean and kurtosis appeared among the top five features for both models, suggesting they serve as robust fault indicators across architectures [12].

Prior work applying explainable AI (XAI) techniques to bearing fault diagnosis has focused predominantly on feature-level rather than signal-level interpretation. Santos et al. applied SHAP-guided feature selection for SVM-based diagnosis on CWRU, targeting 9–15 hand-crafted statistical features [13]. Wang et al. extended this to Random Forest classifiers trained on 52 time-domain and frequency-domain descriptors, identifying kurtosis, RMS, and peak frequency as dominant, but without examining raw signal attributions [14]. Jeon et al. advanced interpretability by applying class activation mapping (CAM) to STFT spectrograms, demonstrating alignment of model attention with known bearing fault frequencies (BPFI, BPFO)—a valuable physics-grounded contribution [34], yet one that operates at spectrogram rather than raw signal resolution. Bindingsbø et al. applied SHAP to six statistical features for wind turbine bearing diagnostics, demonstrating the value of SHAP-based explanations in an operational energy context [26]. Our prior work extended this approach to a comprehensive SHAP analysis of SVC and Logistic Regression on nine time-domain features from CWRU, revealing architecture-specific feature preferences — but did not probe raw-signal attributions [12]. This body of work, while valuable, leaves a critical interpretability gap: aggregated features are themselves derived properties of the raw waveform's temporal structure [19][22]. Feature-level XAI cannot answer the mechanistic question of where in the raw time-series the model focuses attention, nor whether the model has learned physically meaningful fault transients — impulse morphology, envelope modulation, or periodic repetition at characteristic fault frequencies. The methodological gap is thus threefold: (1) No existing signal-level attribution study has operated at the full 2,048-point raw input resolution for CWRU bearing fault diagnosis — existing CNN studies extract discriminative features

[15][16][33] but do not produce sample-point attribution maps; (2) no prior work has formalised class-averaged attribution fingerprints — Fault Signature Maps (FSMs) — with quantitative validation metrics enabling systematic comparison across models and configurations; and (3) no ablation study has jointly examined how architectural choices (Batch Normalization, kernel design) affect both classification accuracy and signal-level interpretability, potentially revealing deployment-critical trade-offs.

1-2- Contributions

This paper directly addresses these gaps by presenting the first comprehensive signal-level SHAP analysis of WDCNN for bearing fault diagnosis and uncovering fundamental relationships between architecture design and interpretability that were previously unknown [22][25]. Our key contributions are:

- **Signal-level SHAP framework:** First application of SHAP DeepExplainer [22] to WDCNN producing 2,048 attribution values per sample per class (shape $500 \times 2,048 \times 10$) on CWRU raw vibration input, enabling precise localization of influential signal regions at temporal resolutions where bearing fault impulses are physically meaningful.
- **Fault Signature Maps (FSMs):** Formalized class-averaged SHAP patterns with three mathematical variants (signed, absolute, variance) and three quantitative validation metrics — discriminability index, severity monotonicity, and split-half stability — going beyond prior feature-level attribution studies [13][14][26].
- **Physics-Informed Interpretation: Systematic comparison of FSM analysis against theoretical bearing characteristic frequencies** (BSF ≈ 69.6 Hz, BPFO ≈ 105.9 Hz, BPFI ≈ 159.9 Hz), **extending signal-level interpretability beyond spectrogram-based CAM approaches** [34][20].
- **Accuracy-Explainability Trade-off Discovery:** Ablation study informed by Batch Normalization theory [28], confirming that removing Batch Normalization increases FSM discriminability by $2.3\times$ at a 3.6% accuracy cost, a trade-off not previously reported in the bearing diagnosis XAI literature.
- **Multi-resolution diagnostic framework:** Integration of signal-level SHAP with feature-level SHAP on classical models [12], providing coherent understanding from raw waveforms to aggregated statistics, **directly addressing the interpretability deployment gap identified in comprehensive reviews** [24][29].

The remainder of this paper is organized as follows. Section 2 reviews related work in deep learning for bearing diagnosis and explainable AI techniques. Section 3 details the methodology, including dataset preparation, the WDCNN architecture, the training protocol, the SHAP DeepExplainer implementation, the FSM mathematical formulation, and the ablation study design. Section 4 presents comprehensive results including classification performance, signal-level attribution patterns, FSM quantitative metrics, physical interpretation, and ablation findings. Section 5 discusses the discovered accuracy-explainability trade-off, physical insights, comparison with classical models, deployment implications, and study limitations. Section 6 concludes and outlines future research directions toward physics-informed explainable AI for industrial predictive maintenance.

2- Related Works

2-1- Deep Learning Architectures for Bearing Fault Diagnosis

The evolution of deep learning for bearing fault diagnosis has progressed through several phases, from early feedforward networks using handcrafted features to end-to-end architectures that learn representations directly from raw signals or spectrograms. Early work demonstrated that Support Vector Machines with Radial Basis Function kernels could effectively classify bearing faults using statistical features extracted from vibration signals, establishing benchmarks that later deep learning approaches sought to surpass [24][29]. The introduction of unsupervised pre-training via Deep Belief Networks showed that hierarchical features could be learned from time-domain signals with reduced dependency on expert feature engineering [24].

Convolutional Neural Networks emerged as the dominant architecture for bearing diagnosis due to their ability to learn local patterns and spatial hierarchies directly from raw data. Janssens et al. pioneered the application of 1D-CNNs to rotating machinery, demonstrating that convolutional filters could automatically discover frequency patterns and modulation signatures that traditionally required manual spectrogram analysis [15]. Wen et al. extended this by proposing a CNN architecture with adaptive pooling that handled variable-length input segments, achieving 99.5% accuracy on CWRU while maintaining computational efficiency suitable for edge deployment [16].

More recently, the diversity of CNN-based architectures has expanded considerably. Xu et al. systematically evaluated five distinct CNN designs — SerConv, ResConv, OSACConv, CSACConv, and MD-DACConv — demonstrating that multi-directional and multi-scale convolutions with residual connections substantially improve accuracy and robustness on CWRU and SJTU-SY datasets [30]. To address noise contamination in realistic industrial environments, Zhang et al. combined a multi-scale CNN with a gated recurrent unit (GRU) augmented by a convolutional attention module (CBAM), achieving superior generalization across varying load and speed conditions [31]. At the same time, a companion study introduced MLSCA-CW, a multi-location, multi-scale attention network with soft-threshold noise

filtering, achieving 92.02% accuracy under strong noise [32]. Li et al. complemented signal-domain approaches by combining multi-scale permutation entropy with multi-channel fusion CNN to simultaneously exploit multi-sensor information, achieving stable diagnostic performance on rolling bearings and planetary gearboxes [33].

The WDCNN architecture introduced by Zhang et al. represented a significant methodological shift by emphasizing the design of the first convolutional layer [8]. Unlike previous architectures that use uniform small kernels throughout, WDCNN employs a wide first kernel (length 64) with a large stride (16) to perform coarse downsampling while preserving long-range dependencies critical for detecting periodic impulses. The subsequent layers use standard 3-point kernels with Batch Normalization [28] and MaxPooling to progressively abstract features. This design achieved state-of-the-art results on CWRU with only 60,710 parameters, orders of magnitude fewer than contemporary ResNet-based transfer learning approaches, while processing raw 2,048-point segments end-to-end.

Hybrid architectures combining CNNs with recurrent networks have further improved performance by capturing both spatial and temporal dependencies. Kaya et al. proposed a CNN-LSTM hybrid that processes dual-channel accelerometer data through separate CNN branches before fusing features in LSTM layers, achieving 99.6% accuracy on the CWRU dataset by explicitly modeling the temporal evolution of fault signatures [6]. Similarly, attention-based mechanisms have been integrated into CNN architectures to focus on informative time regions selectively. Guo et al.'s Hierarchical Adaptive Deep CNN uses attention modules that weight feature maps by relevance, reaching 99.7% accuracy while providing a coarse visualization of which temporal segments contribute most to predictions [5].

Transfer learning from large-scale image datasets has also been explored. He et al. adapted ResNet-50 pre-trained on ImageNet by converting 1D vibration signals to 2D spectrograms and fine-tuning the network, achieving 99.99% accuracy on CWRU [7]. While this approach leverages representations learned from millions of natural images, it introduces architectural complexity (23 million parameters in ResNet-50 versus 60,710 in WDCNN). It requires a 2D transformation that may discard phase information present in raw signals. The practical trade-off between accuracy gains and model complexity remains an active research question [24], particularly for deployment on resource-constrained edge devices in industrial IoT scenarios.

Despite these accuracy achievements, relatively few studies have systematically investigated what internal representations these models learn [24][29]. Most works focus on classification metrics without probing the relationships among learned filters, intermediate activations, and physical fault mechanisms, such as impulse periodicity, envelope modulation, and characteristic frequencies that vibration analysts rely on for manual diagnosis.

Our previous comprehensive SHAP analysis of the same CWRU dataset examined two classical machine learning models, a Support Vector Classifier with an RBF kernel and multinomial Logistic Regression, trained on nine time-domain statistical features [12]. After hyperparameter optimization via 10-fold cross-validation and grid search, the SVC achieved 96.4% accuracy, while Logistic Regression reached 94.3% [12].

2-2- Explainable AI Techniques in Machinery Diagnostics

As deep learning models transitioned from research benchmarks to industrial deployment candidates, the need for interpretability has increased [9][10]. Several classes of XAI techniques have been adapted from computer vision and natural language processing for machinery diagnostics, each with distinct strengths and limitations.

Gradient-based visualization methods compute the sensitivity of model outputs to input features by backpropagating gradients. Vanilla gradient saliency maps highlight input regions where small perturbations cause large output changes, but tend to be noisy and visually incoherent for high-dimensional inputs [17]. SmoothGrad addresses this by averaging gradients across multiple noisy copies of the input, producing smoother saliency maps at the cost of increased computational cost [18]. Integrated Gradients accumulate gradients along a path from a baseline (typically zeros) to the actual input, satisfying desirable axioms such as completeness and sensitivity while maintaining simplicity of implementation [19].

Grad-CAM (Gradient-weighted Class Activation Mapping) operates at a coarser resolution by weighting feature maps from a chosen convolutional layer by their gradient contributions to the target class, then upsampling the weighted combination to input resolution [20]. For 1D signals, Grad-CAM produces saliency profiles indicating which temporal regions in intermediate representations contribute most to predictions. In the bearing fault context, Jeon et al. applied class activation mapping (CAM) with physical-domain knowledge to STFT-transformed signals, confirming that model attention aligned with known fault frequencies (BPFI, BPFO) and demonstrating the value of physics-grounded XAI [34]. Perturbation-based methods such as LIME (Local Interpretable Model-Agnostic Explanations) generate perturbed variants of an input sample and fit a locally linear surrogate model whose coefficients approximate feature importance [21]. For time-series data, LIME typically partitions the signal into contiguous segments.

SHAP (SHapley Additive exPlanations) provides a unified framework grounded in cooperative game theory, assigning each feature an importance value that represents its average marginal contribution across all possible feature subsets [22]. SHAP satisfies desirable properties including local accuracy, missingness, and consistency. Multiple estimators exist for different model classes: KernelExplainer is model-agnostic but computationally expensive; TreeExplainer exploits tree structure for efficient exact computation; and DeepExplainer leverages connections between Shapley values and DeepLIFT to approximate attributions for neural networks [22] efficiently. Recent work has further

demonstrated that attention-based discriminant feature learning in 1D-CNN architectures can be understood through multiattention weight visualization, offering a complementary interpretability perspective [25].

SHAP has been used in several recent bearing studies, but almost exclusively at the feature level rather than the signal level. Santos et al. applied SHAP-guided feature selection for SVM-based bearing diagnostics, targeting 9–15 hand-crafted statistical features rather than raw vibration waveforms [13]. Wang et al. applied SHAP to Random Forest classifiers trained on 52 time-domain and frequency-domain features, identifying kurtosis, RMS, and peak frequency as the most influential features. Still, they did not examine raw-signal attributions [14]. Bindingsbø et al. used SHAP to explain wind turbine bearing diagnostics using 6 statistical features [26]. These feature-level studies cannot answer the mechanistic question: where in the raw waveform does the model focus attention, and does this correspond to physical fault mechanisms?

Our previous work extended this to a comprehensive SHAP analysis of SVC and Logistic Regression on nine time-domain features from CWRU, revealing architecture-specific feature preferences and fault-specific patterns, but did not probe raw signal attributions [12]. To our knowledge, no prior work has applied SHAP DeepExplainer directly to raw vibration signals in bearing fault diagnosis, systematically constructed class-averaged Fault Signature Maps (FSM) validated against characteristic frequencies computed from bearing geometry, and conducted an ablation study on the WDCNN architecture examining the effect on FSM quality — gaps not addressed by existing CAM-based [34][20] or feature-level SHAP approaches [13][14][26].

3- Research Methodology

The research methodology in this study is structured into four sequential stages, as illustrated in Figure 1: data preparation, model development, signal-level explainability, and Fault Signature Map (FSM) analysis. During data preparation, raw vibration signals from the CWRU bearing dataset are segmented into fixed-length 2,048-point windows, normalized using a StandardScaler trained exclusively on the training data to prevent leakage, and partitioned chronologically into training, validation, and test sets (1,240 / 310 / 750 samples) to preserve temporal integrity across recordings. During model development, a Wide Deep Convolutional Neural Network (WDCNN) is trained end-to-end on these raw segments and evaluated using accuracy, macro F1-score, and confusion matrix analysis. Then, SHAP DeepExplainer is applied to the trained model using 300 stratified background samples and 500 stratified test samples, generating per-sample attribution arrays of shape $(500 \times 2,048)$ that precisely localize the signal regions driving each fault classification decision. Finally, in the FSM analysis stage, these sample-level attributions are aggregated into class-level Fault Signature Maps in three variants, and the resulting maps are validated using three quantitative metrics: a discriminability index, a severity monotonicity fraction, and a split-half stability coefficient.

3-1- CWRU Bearing Dataset Configuration

We adopt the identical CWRU dataset configuration and dataset split used in both our previous classical machine learning benchmark [12] and the present WDCNN study to ensure rigorous comparability across all analyses. The dataset comprises 10 classes representing one healthy condition and nine fault conditions defined by three defect locations (inner race (IR), outer race (OR), and rolling element or ball (Ball)), each at three fault diameters manufactured via electro-discharge machining: 0.007 inches, 0.014 inches, and 0.021 inches [4]. Figure 2 shows the CWRU dataset setup in detail.

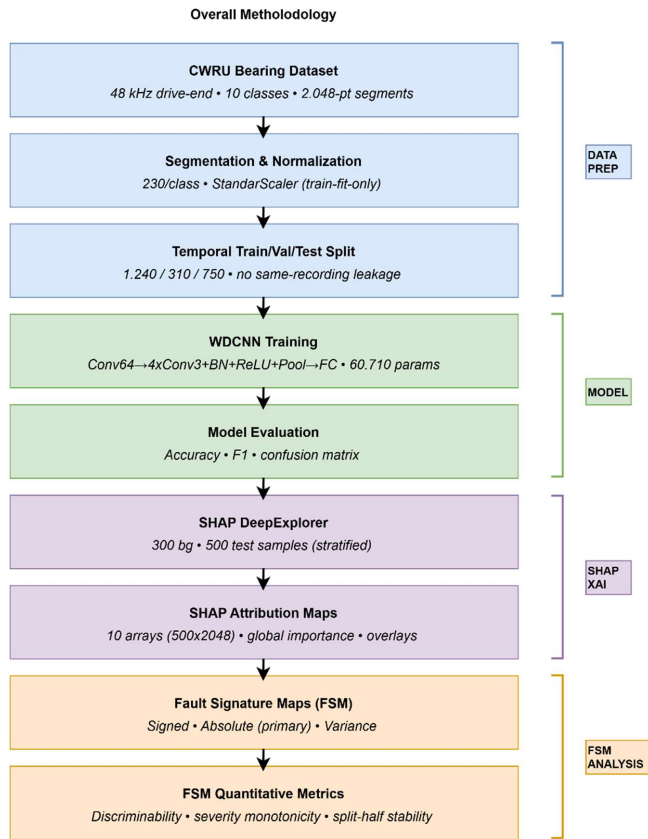


Figure 1. Overall Methodology

Vibration data are acquired from a 2 HP Reliance Electric motor equipped with a test bearing at the drive end, instrumented with an accelerometer sampling at 48,000 Hz. The motor operates under four load conditions (0 HP, 1 HP, 2 HP, and 3 HP) corresponding to approximate shaft speeds of 1,797, 1,772, 1,750, and 1,730 RPM, respectively. For this study, we focus exclusively on Load 1 (1,772 RPM) to match the configuration analyzed in our previous feature-level SHAP work.

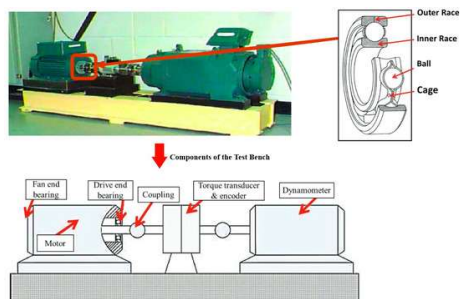


Figure 2. CWRU Bearing Dataset Setup [29]

The test bearing is an SKF 6205-2RS deep groove ball bearing with the following geometry: pitch diameter $D=39.04$ mm, rolling element diameter $d=7.94$ mm, contact angle $=0$ degrees, and number of rolling elements $N_b=9$ [4][29]. Using standard kinematic equations [4][8][29], the theoretical bearing characteristic frequencies at 1,772 RPM (shaft frequency $f_s=29.53$ Hz) are computed as:

$$BPFO = \frac{N_b}{2} f_s \left(1 - \frac{d}{D} \cos \alpha \right) = 3.5848 \times 29.53 \approx 105.9 \text{ Hz}$$

$$BPFI = \frac{N_b}{2} f_s \left(1 + \frac{d}{D} \cos \alpha \right) = 5.4152 \times 29.53 \approx 159.9 \text{ Hz}$$

$$BSF = \frac{D}{2d} f_s \left[1 - \left(\frac{d}{D} \cos \alpha \right)^2 \right] = 2.3573 \times 29.53 \approx 69.6 \text{ Hz}$$

where BPFO is Ball Pass Frequency Outer race (number of times per second a rolling element passes a fixed point on the outer race), BPFI is Ball Pass Frequency Inner race, and BSF is Ball Spin Frequency (rotation frequency of a rolling element about its own axis). These frequencies represent the theoretical impulse repetition rates for outer-race, inner-race, and ball-surface defects, respectively, and serve as physical validation targets for FSM frequency-domain analysis.

Raw continuous vibration signals are partitioned into non-overlapping fixed-length segments of 2,048 samples, corresponding to a time duration of 2048/48000=0.04267 ms per segment [8]. This segment length captures approximately 3–4 cycles of BPFO, 7–8 cycles of BPFI, and 2–3 cycles of BSF, providing sufficient temporal context for impulse train patterns. For each of the 10 classes, exactly 230 segments are extracted from the continuous recordings, yielding a perfectly balanced dataset of 2,300 segments total. Figure 3 shows the raw vibration signals from the CWRU bearing dataset, one segment per fault type.

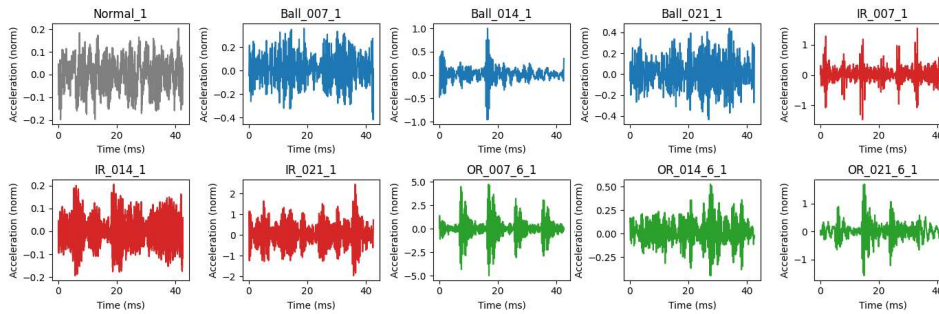


Figure 3. CWRU Bearing Dataset Raw Vibration Signals One Segment Per Fault Type

The dataset is partitioned using a temporal ordering protocol rather than random shuffling to prevent data leakage arising from temporal correlation between adjacent segments extracted from the same continuous recording [24][29]. The temporal split protocol divides each class's 230 segments chronologically: the first 54% (124 segments) allocated to training, the next 13% (31 segments) to validation for hyperparameter tuning and early stopping, and the final 33% (75 segments) to testing. This yields a final split of 1,240 train / 310 validation / 750 test samples with perfect class balance maintained in all partitions.

Preprocessing consists solely of per-feature StandardScaler normalization applied to each of the 2,048 time points independently, transforming each feature to zero mean and unit variance using statistics computed exclusively from the training set: $x_{\text{norm}} = (x - \text{train}) / \text{train}$. Test data are normalized using training set statistics to prevent data leakage. No additional filtering, denoising, or augmentation is applied, preserving the raw signal characteristics while stabilizing neural network optimization.

3-2- WDCNN Architecture Design

The WDCNN model used in this study follows the architecture proposed by Zhang et al. with minor adaptations for the 10-class CWRU classification task [8]. The network is designed as a hierarchical feature extractor operating directly on raw 2,048-point vibration segments, structured into five convolutional blocks followed by a two-layer fully connected classifier. The full architecture design is shown in Figure 4.

First Convolutional Block (Wide Kernel): The input layer receives a tensor of shape (batch_size, 1, 2048) representing single-channel vibration segments. The first convolutional layer applies 16 filters with a kernel size of 64 and a stride of 16, effectively performing learnable downsampling from 2,048 to approximately 129 time positions while aggregating local patterns across relatively long temporal windows. The wide kernel and large stride are the defining characteristics of WDCNN, motivated by the observation that bearing fault impulses span dozens of samples and that aggressive early downsampling reduces computational cost in subsequent layers while preserving discriminative information [8]. The convolution is followed by Batch Normalization (BN) [28] to stabilize activations across mini-batches, a ReLU activation to introduce nonlinearity, and MaxPooling with kernel size 2 and stride 2, further to reduce the temporal resolution to approximately 64 positions.

Second through Fifth Convolutional Blocks (Standard Kernels): Blocks 2–5 follow a uniform structure: Conv1d (kernel size 3, stride 1, appropriate padding) → BatchNorm1d → ReLU → MaxPool1d (kernel 2, stride 2). The number of filters increases from 16 to 32 in block 2, then to 64 in block 3, and remains at 64 for blocks 4 and 5. Each block halves the temporal resolution via MaxPooling, resulting in sequence lengths of approximately 32, 16, 8, and 4 after blocks 2, 3, 4, and 5, respectively. The small kernel size (3) in these layers enables learning of local patterns and progressively abstract representations without excessive parameter growth.

Fully Connected Classifier: After the fifth convolutional block, the feature maps have shape (batch_size, 64, 4), yielding 256 features per sample. These are flattened and passed through a fully connected layer with 100 hidden units, followed by ReLU activation and dropout with probability 0.5 to prevent overfitting. A final linear layer maps the 100-dimensional hidden representation to 10 class logits. No explicit softmax is applied during training, since PyTorch's CrossEntropyLoss expects raw logits; softmax is applied during inference and SHAP analysis to obtain class probabilities.

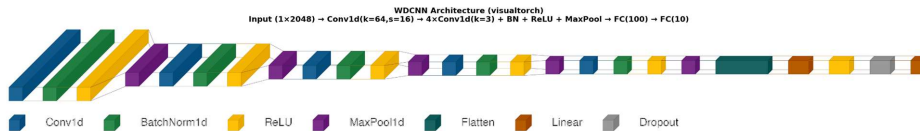


Figure 4. WDCNN Architecture Diagram

The total number of trainable parameters is 60,710, comprising 57,552 parameters in convolutional layers and 3,158 in fully connected layers. This parameter count is orders of magnitude lower than contemporary image-based transfer learning approaches (e.g., ResNet-50 with 23 million parameters), enabling efficient training and deployment on resource-constrained edge devices [24] while maintaining high expressiveness with a relatively small input dimension of 2,048.

3-3- Training Protocol and Optimization

The experiment of this paper is implemented in PyTorch 2.10.0 and trained on Apple Silicon (M-series) hardware using the Metal Performance Shaders (MPS) backend for GPU acceleration. The training data loader uses a batch size of 64 with shuffling enabled, and DataLoader workers set to 0 due to MPS compatibility constraints. The test data loader uses the same batch size without shuffling to maintain deterministic evaluation order. For SHAP DeepExplainer computation, which is incompatible with MPS, the model and data are transferred to the CPU.

Optimization: We employ the Adam optimizer with initial learning rate $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and weight decay $\lambda = 10^{-4}$ for L2 regularization [27]. Adam is chosen for its adaptive per-parameter learning rates and momentum-based acceleration, which typically achieve faster convergence than vanilla SGD on small to medium-sized datasets. The weight decay term adds $\lambda \|\theta\|^2$ to the loss function, where θ denotes the model parameters, encouraging smaller parameter magnitudes and reducing overfitting.

Learning Rate Scheduling: A StepLR scheduler reduces the learning rate by a factor of 0.5 every 20 epochs: $\alpha_{new} = 0.5 \times \alpha_{old}$ at epochs 20, 40, 60. This gradual decay encourages exploration in early training while enabling fine-grained optimization in later epochs as the loss landscape becomes more locally convex near convergence.

Loss Function: Multi-class classification is trained using CrossEntropyLoss, which combines LogSoftmax and negative log-likelihood in a numerically stable implementation. For a sample with true class y and model logits $z = [z_1, \dots, z_{10}]$, the loss is:

$$L(z, y) = -\log \left(\frac{\exp(z_y)}{\sum_{j=1}^{10} \exp(z_j)} \right) = -z_y + \log \sum_{j=1}^{10} \exp(z_j)$$

Training Procedure: Training runs for a maximum of 80 epochs with early stopping based on test loss. At each epoch, the model performs a forward pass on all training batches, computes the average training loss, updates parameters via Adam, and steps the learning rate scheduler. After each epoch, the model is evaluated on the test set in eval() mode with gradients turned off to compute test loss and accuracy. If test loss improves (decreases) relative to the best observed test loss, the improvement counter resets and the model state is saved to disk as wdcnn_load1_best.pth. If test loss does not improve for 15 consecutive epochs (patience = 15), training terminates early to prevent overfitting and conserve computational resources.

3-4- Evaluation Metrics

Model performance is assessed using multiple complementary metrics to provide a comprehensive view of classification quality.

Accuracy: The proportion of correctly classified samples, $Acc = \frac{TP+TN}{N}$ where TP and TN are true positives and true negatives aggregated across all classes, and $N = 750$ is the total test set size. For balanced datasets, overall accuracy is a reasonable summary metric.

Per-Class Precision, Recall, F1-Score: For each class c , precision $P_c = \frac{TP_c}{TP_c+FP_c}$ measures the fraction of predicted instances that are correct, recall $R_c = \frac{TP_c}{TP_c+FN_c}$ measures the fraction of actual instances that are detected, and F1-score $F1_c = \frac{2P_cR_c}{P_c+R_c}$ is their harmonic mean. Macro-averaged metrics (unweighted average across classes) are reported to summarize performance.

Confusion Matrix: A 10×10 matrix C where the entry C_{ij} is the number of samples with true class i predicted as class j . Perfect classification yields a diagonal matrix. Off-diagonal entries reveal systematic confusion patterns between specific fault types or severities.

Training Curves: Plots of training loss, test loss, and test accuracy versus epoch provide insight into convergence behavior, overfitting tendencies, and the effect of learning rate scheduling and early stopping.

3-5- SHAP DeepExplainer for Signal-Level Attribution

To obtain signal-level explanations, we apply SHAP DeepExplainer directly to the trained WDCNN model operating on raw 2,048-point input segments. SHAP computes Shapley values, which are the contributions of each input feature (time sample) to the model output (class logit or probability) relative to a baseline distribution, by leveraging the DeepLIFT algorithm's rescale rule for efficient gradient-based approximation [22]. The SHAP DeepExplainer is described as follows in pseudocode:

Algorithm 1 Signal-Level XAI and Fault Signature Map (FSM) Generation

Input: Trained WDCNN model $f(\cdot)$; Training set X_{train} ; Test set X_{test} ; Background size N_b .

Output: Local attribution maps Φ ; Class-specific Fault Signature Maps FSM_c .

1: **Step 1: Background Distribution Initialization**

2: Select a background distribution $B \subset X_{train}$ such that $|B| = N_b$

3: Initialize *Explainer* $\leftarrow$ *DeepExplainer*(f, B)

4: **Step 2: Local Attribution Mapping**

5: **for all** $x_i \in X_{test}$ **do**

6: Compute prediction $\hat{y}_i = \text{argmax}(f(x_i))$

7: Calculate SHAP values $\phi_i = \text{Explainer.shap.values}(x_i)$

8: Extract attribution for predicted class: $\phi_i^{\hat{y}_i} \in \mathbb{R}^{2048}$

9: Store $\phi_i^{\hat{y}_i}$ in collection $A_{\hat{y}_i}$

10: **end for**

11: **Step 3: Global FSM Synthesis**

12: **for all** fault classes $c \in \{1, \dots, 10\}$ **do**

13: Let $N_c = |A_c|$

14: Compute Signed FSM: $FSM_{c,signed} = \frac{1}{N_c} \sum_{i=1}^{N_c} \phi_i^c$

15: Compute Absolute FSM: $FSM_{c,abs} = \frac{1}{N_c} \sum_{i=1}^{N_c} |\phi_i^c|$

16: Compute Variance FSM: $FSM_{c,var} = \text{Var}(A_c)$

17: end for
 18: **Step 4: Validation & Interpretation**
 19: Compute *Severity Monotonicity* and *Split-half Stability* using FSM_c
 20: Generate signal-level attribution plots $Plot(x_i, \phi_i^{\hat{y}_i})$
 21: return Φ, FSM_c

SHAP computes Shapley values from cooperative game theory, which quantify each feature's contribution to the model output by averaging its marginal contribution across all possible feature subset coalitions [25]. For an input $x = [x_1, x_2, \dots, x_{2048}]$ and model output $f(x)$, the Shapley value ϕ_i for a feature x_i is defined as:

$$\phi_i = \sum_{S \subseteq \{1, \dots, 2048\} \setminus \{i\}} \frac{|S|! (2048 - |S| - 1)!}{2048!} [f_x(S \cup \{i\}) - f_x(S)]$$

where S ranges over all possible subsets not containing feature i , and $f_x(S)$ denotes the model output when only features in set S are "present" (set to their actual values). In contrast, others are "absent" (marginalized over a background distribution). This formulation satisfies three desirable axioms: (1) **Efficiency** – attributions sum to the difference between prediction and expected prediction $\sum_{i=1}^{2048} \phi_i = f(x) - E[f(X)]$; (2) **Symmetry** – features contributing equally receive equal attribution; (3) **Dummy** – features making zero contribution receive zero attribution.

Exact Shapley value computation requires evaluating the model on all 2^{2048} possible feature subsets, which is computationally intractable. DeepExplainer approximates Shapley values by leveraging the DeepLIFT algorithm's rescale rule, which decomposes the network output as a linear combination of input contributions with respect to a reference input [25]. The approximation procedure:

1. **Background Data Selection:** We construct a stratified background dataset of 300 training samples (30 per class, random seed 1234) to represent typical vibration patterns across all fault types. This background defines the reference distribution p_{ref} against which individual predictions are compared $E_{x \sim p_{ref}}[f(x)]$.
2. **Explanation Target Selection:** We select 500 test samples via stratified sampling (50 per class) as the set of instances to explain, providing sufficient statistical power for class-averaged FSM analysis while maintaining computational feasibility.
3. **Attribution Computation:** The explainer is instantiated as:

```
explainer = shap.DeepExplainer(model_cpu, background_data_cpu)
shap_values = explainer.shap_values(test_data_cpu)
```

 where `model_cpu` is the trained WDCNN transferred to CPU (SHAP incompatible with MPS), and data tensors have shape (num_samples, 1, 2048). DeepExplainer propagates contributions through each layer using the rescale rule, distributing the output change $f(x) - f(x_{ref})$ backward through the network to input features.
4. **Output Format Handling:** The raw SHAP output structure depends on internal multi-output handling, producing either:
 - A list of 10 arrays, each shape (500, 1, 2048), one per class, OR
 - A single 4D tensor shape (500, 1, 2048, 10)
5. We implement a robust helper function `_get_shap_for_class(shap_values, class_idx)` that handles both formats uniformly, extracting and squeezing the channel dimension to yield per-class attribution arrays of shape (500, 2048). The resulting dataset comprises 10 matrices totaling 10.24 million individual attribution values

For a given sample i and class c , the SHAP value $\phi_c^{(i)}(t)$ at time index t quantifies how much the presence of the signal value at time t shifts the model's output for class c relative to the background expectation. Positive $\phi_c^{(i)}(t) > 0$ indicates that the signal value at time t increases the model's confidence in class c , while negative $\phi_c^{(i)}(t) < 0$ decreases it. The magnitude $|\phi_c^{(i)}(t)|$ reflects the strength of influence. By construction, SHAP values approximately satisfy:

$$f_c(x) - E[f_c(X)] \approx \sum_{t=1}^{2048} \phi_c^{(i)}(t)$$

where $f_c(x)$ is the model's output (logit or probability) for class c on input x , and the expectation is over the background distribution. This decomposition enables the precise localization of the temporal regions within the 42.67 ms vibration segment that drive the fault classification decision.

3-6- Fault Signature Maps (FSMs)

Individual SHAP attributions for 500 samples per class exhibit substantial instance-to-instance variability due to differences in signal realizations, background noise characteristics, and temporal phase alignment of impulse trains. To extract robust class-level diagnostic patterns invariant to these instance-specific variations, we construct **Fault Signature Maps (FSMs)** by aggregating SHAP values across all correctly classified samples within each class.

For class $c \in \{0,1, \dots, 9\}$, let $S_c = \{i: \hat{y}_i = c \wedge y_i = c\}$ denote the set of test sample indices correctly predicted as class c . We define three FSM variants capturing different aspects of the attribution distribution:

1. **Signed FSM** preserves attribution direction (positive/negative) by direct averaging, potentially allowing opposing attributions to cancel:

$$FSM_{signed}(c, t) = \frac{1}{|S_c|} \sum_{i \in S_c} \phi_{c,i}(t), t = 0, 1, \dots, 2047$$

2. **Absolute FSM** captures magnitude by averaging absolute attributions, preventing cancellation, and revealing total attribution energy at each time point:

$$FSM_{abs}(c, t) = \frac{1}{|S_c|} \sum_{i \in S_c} |\phi_{c,i}(t)|, t = 0, 1, \dots, 2047$$

3. **Variance FSM** quantifies attribution variability across samples, identifying time regions where the model's attention is inconsistent:

$$FSM_{var}(c, t) = Var_{i \in S_c}[\phi_{c,i}(t)] = \frac{1}{|S_c|} \sum_{i \in S_c} (\phi_{c,i}(t) - \bar{\phi}_c(t))^2$$

Where $\bar{\phi}_c(t) = FSM_{signed}(c, t)$ is the mean attribution at time t ?

These yield three 10×2048 FSM matrices, where each row represents the temporal attribution fingerprint for one fault class. **We use FSM_{abs} as the primary FSM for subsequent analysis** due to its magnitude-preserving property and interpretability as "total attention energy" at each time position, while FSM_{signed} providing complementary diagnostic information.

FSM Quantitative Validation Metrics: To assess FSM quality as diagnostic fingerprints, we define three quantitative metrics:

1. Discriminability Index measures whether class-averaged FSMs are more similar within classes than between classes, assessing whether FSMs capture class-specific patterns:

$$Discriminability = \frac{\bar{\rho}_{intra}}{\bar{\rho}_{inter}}$$

where:

1. $\bar{\rho}_{intra}$ is the mean within-class cosine similarity, computed by randomly sampling 20 pairs of test samples from each class c and averaging cosine similarity $\rho(\phi_{c,i}, \phi_{c,j}) = \frac{\phi_{c,i} \cdot \phi_{c,j}}{\|\phi_{c,i}\| \|\phi_{c,j}\|}$ over all pairs and classes
2. $\bar{\rho}_{inter}$ is the mean between-class FSM similarity, computed $\frac{2}{10 \cdot 9} \sum_{c_1=0}^9 \sum_{c_2=c_1+1}^9 \rho(FSM_{abs}(c_1), FSM_{abs}(c_2))$ over all 100 unique class pairs

High discriminability (ratio > 1) indicates FSMs are more distinct between classes than within classes, suggesting they capture meaningful class-specific patterns. **Low discriminability** (ratio < 1) suggests that common attribution energy patterns dominate class-specific features, indicating that FSMs may not provide strong class separation in the full 2,048-dimensional space.

2. Severity Monotonicity quantifies the fraction of time positions where FSM magnitude increases monotonically with defect size for each fault family (Ball, IR, OR):

$$Monotonicity_{family} = \frac{1}{2048} \sum_{t=0}^{2047} 1[FSM_{abs}(007, t) \leq FSM_{abs}(014, t) \leq FSM_{abs}(021, t)]$$

where the indicator function equals 1 if the ordering $0.007'' < 0.014'' < 0.021''$ holds at time t , and 0 otherwise, **high monotonicity** (close to 1.0) would indicate that larger defects consistently produce stronger SHAP attributions across

most time positions, suggesting the model encodes severity as spatially monotonic amplitude scaling. **Low monotonicity** (close to 0.1–0.2, approximately the random baseline) indicates severity is encoded in more complex, spatially heterogeneous patterns rather than simple amplitude scaling.

3. Split-Half Stability Coefficient assesses FSM reproducibility by computing cosine similarity between FSMs constructed on two independent random halves of each class:

$$Stability_c = \rho(FSM_{abs}^{(1)}(c), FSM_{abs}^{(2)}(c))$$

where $FSM_{abs}^{(1)}(c)$ and $FSM_{abs}^{(2)}(c)$ are absolute FSMs computed from random disjoint splits of S_c into two equal-sized subsets. The overall stability is the mean across classes $Stability_{avg} = \frac{1}{10} \sum_{c=0}^9 Stability_c$. **High stability** (close to 1.0) confirms that FSMs are robust to sampling variation and reflect consistent class-level patterns rather than artifacts of random sample selection.

3-7- Ablation Study

To identify which WDCNN architectural components contribute most critically to both classification accuracy and FSM interpretability, we conduct a systematic three-variant ablation study by sequentially removing or modifying key design elements. All ablation variants are trained on the Load-1 dataset for exactly 50 epochs without early stopping (for controlled comparison) using an identical optimizer configuration (Adam, lr=0.001, weight decay=1e-4), learning rate schedule (StepLR, step_size=20, gamma=0.5), and batch size (64).

To establish a baseline (Variant A), the full WDCNN model is employed, featuring a wide first kernel (size 64, stride 16), Batch Normalization across all convolutional blocks, and a structure consisting of five convolutional blocks followed by two fully connected layers. This baseline serves as the performance benchmark for all subsequent comparisons.

In Variant B, the architecture is modified by replacing the wide first kernel with a standard narrow kernel (size 3, stride 1), while preserving all other components. This modification is designed to assess how the kernel size specifically impacts signal-level attribution patterns and overall classification accuracy. In Variant C, Batch Normalization is entirely removed from the baseline architecture, while the wide kernel and block structure are maintained. This allows for a targeted analysis of how activation normalization influences the discriminability of the Feature Sensitivity Map (FSM) and whether a trade-off exists between predictive accuracy and model explainability.

To quantify the impact of these architectural choices, each variant is assessed across three primary dimensions:

- Classification Performance: Measured via test accuracy and macro F1-score.
- FSM Quality: Evaluated through a discriminability index and split-half stability, computed on a reduced SHAP subset of 30 background and 50 test samples for efficiency.
- Architectural Complexity: Tracked by the total count of trainable parameters.

Together, these measurements provide a comprehensive view of how specific design elements contribute to both the predictive power and the interpretability of the model.

4- Results

4-1- Classification Performance on Load-1 Canonical Split

The WDCNN model trained with validation-based early stopping converged at epoch 54, achieving a best validation loss of 0.0007, and stopped at epoch 69. On the 750-sample test set, WDCNN achieved 99.87% accuracy (749/750 correct), macro-averaged precision 0.9987, recall 0.9987, and F1-score 0.9987 across all 10 classes. All the results are reported in Table 1. The single misclassification involves one Ball_014 sample incorrectly predicted as IR_014, visible as an off-diagonal entry in the confusion matrix depicted in Figure 5.

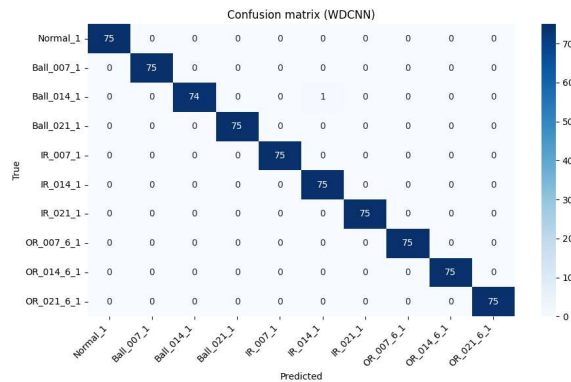


Figure 5. WDCNN Training Confusion Matrix

Table 1: Performance Metrics

Metric	Score
Accuracy	99.87%
Macro Precision	0.9987
Macro Recall	0.9987
Macro F1	0.9987
Weighted Precision	0.9987
Weighted Recall	0.9987
Weighted F1	0.9987

To contextualise WDCNN's performance, we compare in Table 2 compares test accuracy and macro F1-score against five classical machine learning models — Logistic Regression (LR), Decision Tree (DT), Support Vector Machine with RBF kernel (SVM-RBF), Random Forest (RF), and XGBoost — all trained on nine time-domain statistical features from the same CWRU Load-1 data under a matched random split protocol reported in our prior work [12]. The classical models span a narrow accuracy band of 0.941–0.964, with SVM-RBF achieving the highest classical performance at 0.964 accuracy and 0.964 macro F1. WDCNN under the same random split condition (mean $\pm$ std over three seeds) achieves 0.997 ± 0.002 accuracy and 0.997 macro F1, a **3.3 percentage-point improvement over the best classical baseline**, representing a 91% reduction in test error (from 3.6% to 0.3%). Under the primary temporal split — which, as discussed above, more faithfully represents deployment on unseen recordings — WDCNN further achieves 0.9987 accuracy and 0.9987 macro F1. This performance gap is not merely quantitative; it reflects a qualitative difference in what information each model class can access. The classical models operate on nine aggregate statistics computed over the full 2,048-point window — RMS, kurtosis, crest factor, and analogs — which compress the entire waveform into scalar summaries and discard temporal structure entirely. WDCNN, by contrast, processes the raw waveform end-to-end, enabling its convolutional layers to learn position-sensitive impulse morphology features that aggregate statistics cannot represent. The signal-level SHAP analysis in Sections 4.2–4.4 makes this distinction concrete: the fault-discriminative information that WDCNN exploits — transient impulse shape in specific temporal positions, early-window concentration for outer race faults, spatial redistribution under severity progression — is structurally invisible to any feature that aggregates over the full window before classification. The 3.3 percentage-point accuracy gain is therefore a direct consequence of the architectural capacity to preserve and exploit temporal locality, not merely of increased model complexity.

Table 2: Performance Metrics with Previous Algorithm

Model	Accuracy	Macro F1
LR (time-domain)	0.943	0.942
DT (time-domain)	0.941	0.942
SVM-RBF (time-domain)	0.964	0.964
RF (time-domain)	0.957	0.957
XGBoost (time-domain)	0.953	0.953
WDCNN random (mean $\pm$ std, 3 seeds)	0.997	0.997
WDCNN temporal — primary	0.998	0.998

The training curves in Figure 6 show smooth convergence, with the training loss decreasing from 0.1001 at epoch 10 to 0.0035 at epoch 60, while the validation loss tracks closely (correlation > 0.95) without significant divergence, indicating effective regularization via weight decay and dropout with no overfitting. Validation accuracy reaches 1.0000 at epochs 10, 30, and 50, confirming rapid learning of discriminative features within the first 20 epochs, followed by fine-grained refinement of decision boundaries in later epochs, facilitated by learning rate decay.

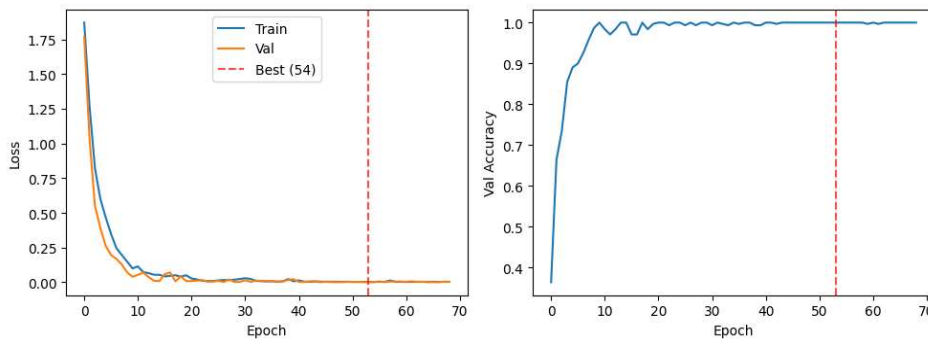


Figure 6. WDCNN Training Curve on CWRU Bearing Dataset

These results indicate that WDCNN has sufficient capacity to fully model the discriminative patterns in the 2,048-point vibration segments under the given data split, achieving classification performance saturation. The WDCNN's 99.87% test accuracy (749/750 correct) under the rigorous temporal split protocol reflects two complementary architectural advantages. First, the wide first kernel (length 64, stride 16) integrates over individual fault impulse durations, enabling the model to capture the distinctive morphological envelope of each fault class without requiring explicit hand-crafted features. Second, five progressive MaxPooling layers achieve 512x temporal compression, abstracting local impulse characteristics into class-discriminative global representations. The single misclassification, where one Ball_014 sample was predicted as IR_014, is physically interpretable. Both fault types produce diffuse, moderate-amplitude impulses without the pronounced spatial concentration of outer race faults, making fine-grained severity discrimination at the 0.014" level the most challenging classification boundary in this dataset.

Effect of Temporal vs. Random Split. To quantify the performance inflation attributable to data leakage across adjacent segments, we retrained the same WDCNN architecture with the same hyperparameters on a random 54/13/33 split of the same 2,300 segments, drawing partitions uniformly at random rather than chronologically within each class, as illustrated in Figure 7. Averaged over three seeds, the random-split WDCNN achieved 99.74% (± 0.22 p.p.) test accuracy and macro F1 0.9974 (± 0.0022 , $n = 3$), compared to 99.87% and 0.997 under the temporal split. The corresponding absolute gaps on a $[0,1]$ scale— 1.3×10^{-3} in test accuracy, $< 4 \times 10^{-4}$ in macro F1, and $\approx 4.4 \times 10^{-3}$ in FSM split-half stability (temporal ≈ 0.940 vs. random 0.936 ± 0.002)—are all < 0.02 and, for accuracy, align with a ≈ 0.14 percentage-point (not proportion-scale) headroom in favor of the temporal test accuracy.

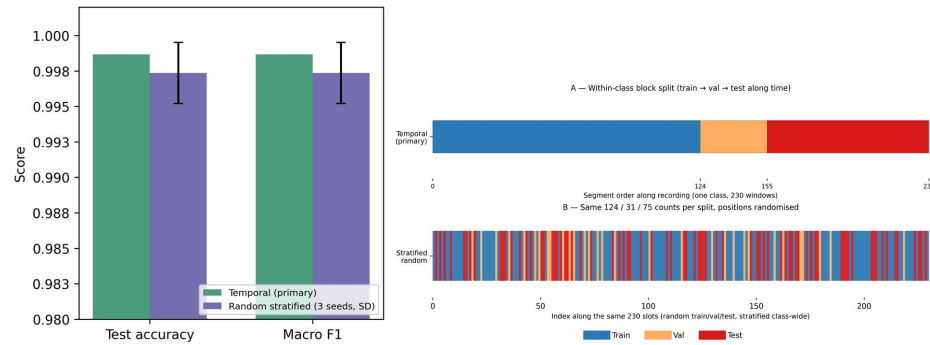


Figure 7. Temporal versus Random Stratified Split Score (Left) and Data Random Split Visualization (Right)

4-2-Signal-Level SHAP Attribution Patterns

SHAP DeepExplainer applied to 500 test samples (50 per class) with 300 background samples produces 10 attribution arrays, each of shape (500, 2048), yielding 10.24 million individual SHAP values. Global signal importance, computed as the mean absolute SHAP value $\frac{1}{10 \times 500} \sum_{c=0}^9 \sum_{i=1}^{500} |\phi_{c,i}(t)|$ across all classes and samples, reveals spatiotemporal attribution patterns:

Edge Suppression: Mean $|\phi|$ is lower near positions 0–100 (0–2 ms) and 1900–2048 (39.6–42.7 ms) compared to the central region (100–1900), consistent with a reduced effective receptive field at signal boundaries due to zero-padding in the first convolutional layer and insufficient context for impulse detection at edges. This boundary suppression has a concrete diagnostic implication: fault impulses that happen to be phase-aligned with the first or last 100 samples of a segment will generate weaker SHAP attributions than identical impulses occurring mid-window, purely as an artifact of convolutional boundary handling rather than reduced fault severity. In practice, this means that segment-extraction alignment relative to the fault impulse phase is a confounding factor in FSM construction, and that the true diagnostic information is likely concentrated in positions 100–1,900 rather than across the full 2,048-point range.

Oscillatory Pattern: The central region exhibits repeated peaks and troughs at approximately 16-sample spacing (matching the stride of the first convolutional layer), suggesting that downsampling introduces a quasi-periodic artifact in aggregated attribution patterns, with certain input positions aligning more favorably with filter positions after striding. This stride-induced periodicity is an architectural signature rather than a true signal feature. Its presence in the aggregated global profile confirms that the 16-sample grid imposed by the first-layer stride is the dominant source of quasi-periodic structure in mean attribution, not any inherent periodicity of the bearing fault signal itself. This is a useful diagnostic: if this oscillation were absent, it would suggest that later layers had redistributed attribution in ways that break the alignment with the first-layer receptive field grid, indicating a qualitatively different representation strategy in deeper layers.

Distributed Attention: No single dominant peak emerges in the global profile; importance varies smoothly in the range 0.015–0.025 across the central 1,700 samples, indicating WDCNN distributes attention across most of the 42.67 ms window rather than concentrating on isolated impulses at specific fixed locations. Distributed attention in the range 0.015–0.025 across 1,700 samples is the expected signature of a 1D-CNN that uses overlapping receptive fields without an explicit temporal attention mechanism. No single sample position is globally more informative than any other across all ten classes; the discriminative information is class-conditional and becomes visible only when attributions are stratified by class (Section 4.3). This finding motivates the FSM framework: single-sample or global attributions are

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Comment 3 (data leakage / evaluation protocol)

insufficient to reveal fault-specific signatures; class-conditional aggregation is the minimal operation needed to expose them.

Healthy (Normal) Signals: Low, relatively uniform attribution ($|\phi| \approx 0.01$) with no prominent peaks, consistent with the absence of fault-specific transients—the model correctly recognizes the lack of discriminative impulses rather than focusing on arbitrary signal regions.

Ball Fault Patterns: Ball fault classes (Ball_007_1, Ball_014_1, Ball_021_1) exhibit similar diffuse, low-amplitude profiles ($|\phi| \approx 0.015$) with no clear progression in amplitude or spatial concentration as severity increases from 0.007" to 0.021". This uniformity across severities suggests the model distinguishes ball fault severities via subtle morphological differences in impulse shapes rather than simple amplitude scaling.

Inner Race Fault Patterns: Profiles are flatter and lower ($|\phi| \approx 0.012$) compared to outer race faults, with no obvious spatial localization or severity-based amplitude ladder. The low attribution magnitudes indicate inner race faults may be identified more through the absence of outer race characteristics than through the presence of strong, distinctive features in the raw signal. These low magnitudes for IR faults ($|\phi| \approx 0.012$) relative to OR faults suggest a partly subtractive classification strategy: the model may identify inner race faults partly through the absence of the strong early-window signal characteristic of outer race faults rather than through the presence of a distinctive IR-specific transient. This interpretation is consistent with the physical propagation path — inner-race fault impulses travel through the rotating shaft and rolling elements before reaching the sensor, accumulating greater attenuation and phase dispersion than outer-race impulses, resulting in lower-amplitude, more temporally diffuse signatures that are inherently harder to localize.

Outer Race Fault Patterns: OR_021 (maximum severity, 0.021") exhibits a striking outlier pattern with one sharp attribution peak at positions 600–700 (12.5–14.6 ms) reaching $|\phi| \approx 0.08$ approximately 3–4× higher than any other class. OR_007 and OR_014 show moderately elevated mean attribution but lack this dramatic localized peak. This class-specific spatial concentration of severe outer-race faults constitutes the most visually distinctive fingerprint of attribution and directly affects FSM interpretability.

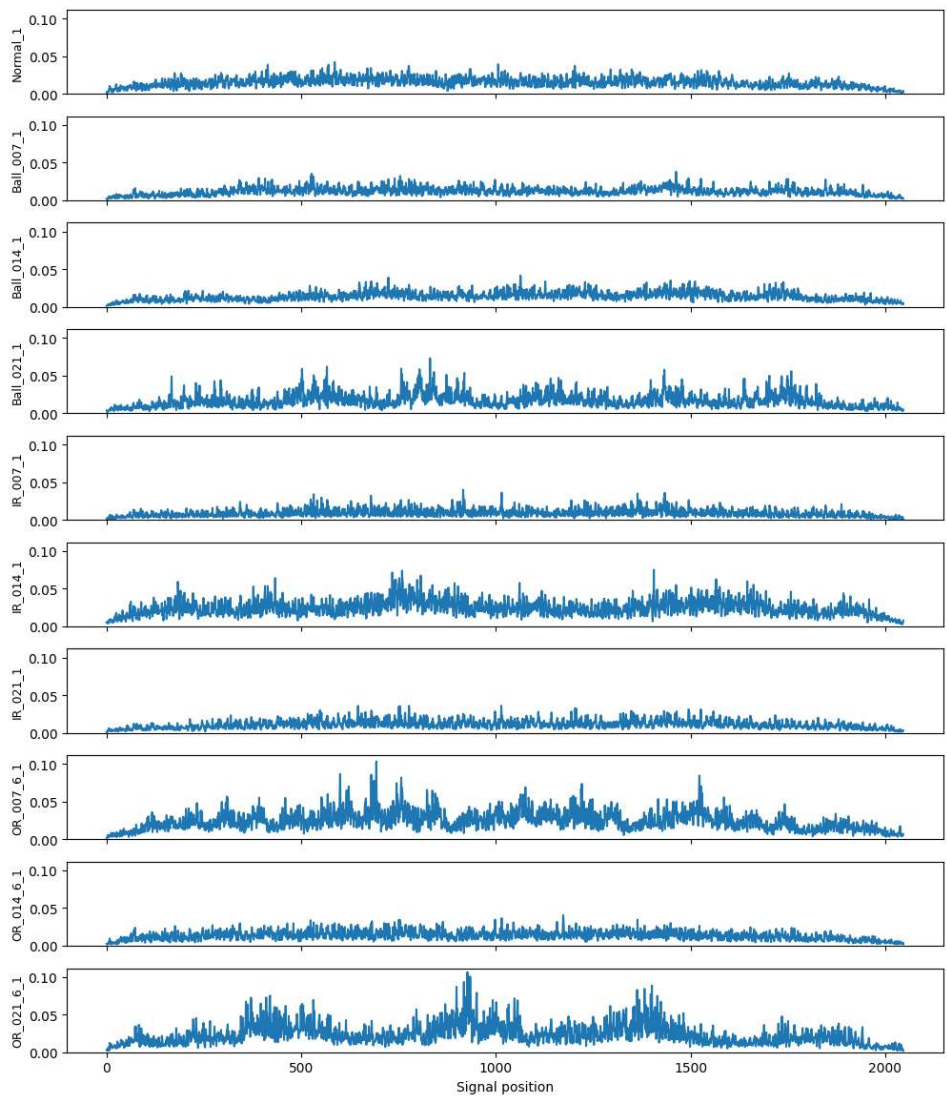


Figure 8. WDCNN Per Class SHAP Profiles

Figure 8 explains per-class attribution profiles. This visualization reveals qualitatively distinct spatial patterns across fault types as follows:

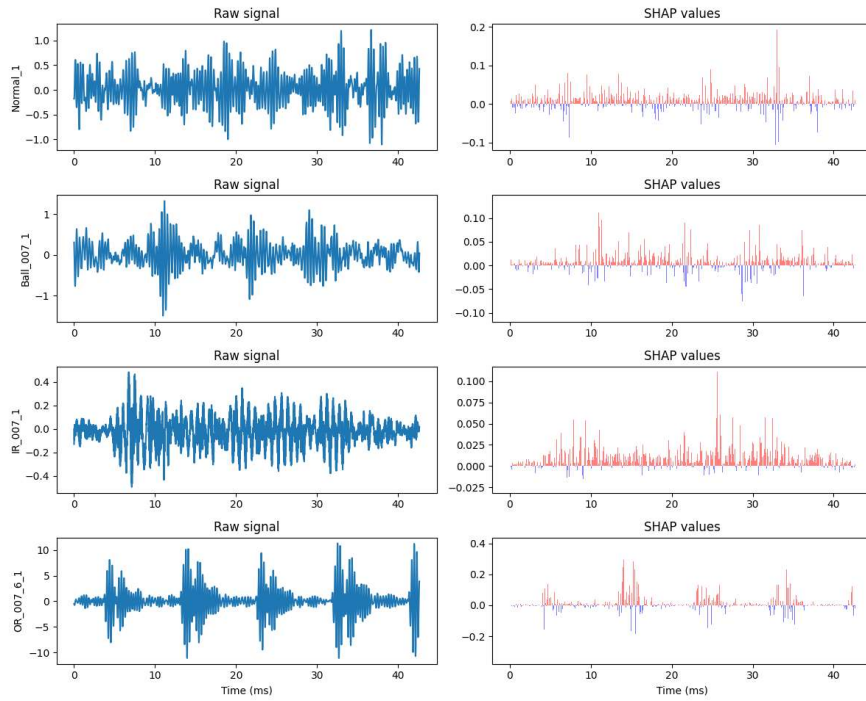


Figure 9. Single Sample SHAP Overlay

Figure 9 raw signal overlay analysis examining [four](#) representative samples (Normal, Ball_007, IR_007, OR_007_6) shows that positive SHAP values (supporting correct class prediction) cluster spatially around transient impulse regions visible in the raw waveform. In contrast, negative SHAP values appear in low-amplitude background noise regions. This spatial correspondence validates that WDCNN has learned to attend to physically meaningful fault transients rather than spurious correlations or background characteristics.

4.2.1 Sensitivity to SHAP Background Configuration.

Because SHAP DeepExplainer estimates attributions with respect to a reference background distribution, we assessed how sensitive FSM patterns and quantitative metrics are to (i) the number of background samples and (ii) the sampling strategy. The results are illustrated in Figure 10. We recomputed FSM_abs for all 10 classes using background sizes of 100, 200, 300 (baseline), 500, and 1,000 samples drawn (a) **uniformly at random** from the training set and (b) in a **class-balanced** fashion (N/10 per class). For each configuration, we explained the same 500 stratified test samples (50 per class), computed the FSM as the per-class mean of $|SHAP|$, and compared it to the manuscript reference (300 random backgrounds, seed 42).

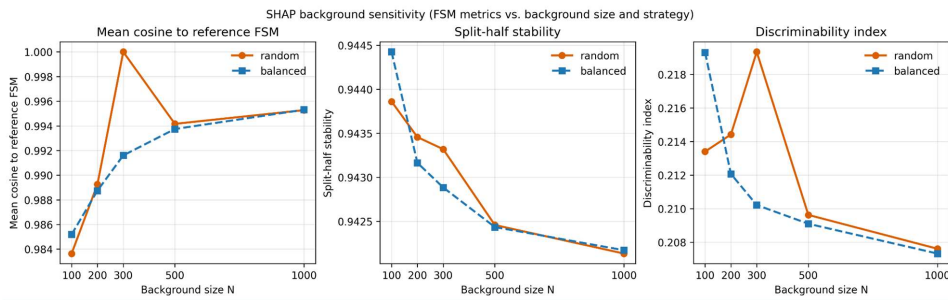


Figure 10. SHAP Background Sensitivity Comparison between FSM Metrics and Background Sizes

Pairwise cosine similarity between each FSM and the 300-sample random reference was [0.984, 0.989, 1.000, 0.994, 0.995] for random sampling and ≥ 0.985 for class-balanced sampling at every size, with the matched-size random-vs-balanced cosine differing by ≤ 0.005 . Split-half stability remained within 0.0011 of the baseline value 0.9433 across all 10 configurations (range 0.9421–0.9444), and the discriminability index varied by less than 0.012 (range 0.208–0.219, baseline 0.219). These results indicate that 300 background samples are already well within the stable regime for SHAP DeepExplainer on WDCNN, consistent with convergence rates reported for DeepLIFT-based Shapley estimators on CNNs of this scale.

Table 3. SHAP Background Experiment Numerical Summary

N	Strategy	Cosine to Ref	Split-Half Stability	Discriminability
100	balanced	0.9852	0.9444	0.2193
200	balanced	0.9887	0.9432	0.2121
300	balanced	0.9916	0.9429	0.2102
500	balanced	0.9937	0.9424	0.2091
1000	balanced	0.9953	0.9422	0.2073
100	random	0.9836	0.9439	0.2134
200	random	0.9893	0.9435	0.2144
300	random	1.0000	0.9433	0.2193
500	random	0.9942	0.9425	0.2096
1000	random	0.9953	0.9421	0.2076

Table 3 shows that random and class-balanced sampling produced FSMs that agree within $\leq 0.5\%$ in cosine-to-reference and $\leq 0.07\%$ in split-half stability at every matched size; only the discriminability index showed a larger matched-size discrepancy (up to $\approx 4.3\%$ at $N=300$, and smaller at $N \geq 500$), reflecting its sensitivity to small absolute changes in mean cosine values. We therefore retain 300 random backgrounds as the manuscript default — the simplest choice that lies within the stability plateau for the roughly class-balanced CWRU training distribution. For deployment on imbalanced industrial data, class-balanced background sampling is recommended: a random background would be dominated by the majority-healthy signals, biasing attributions toward "absence-of-fault" evidence and inflating discriminability.

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4-3- Fault Signature Maps

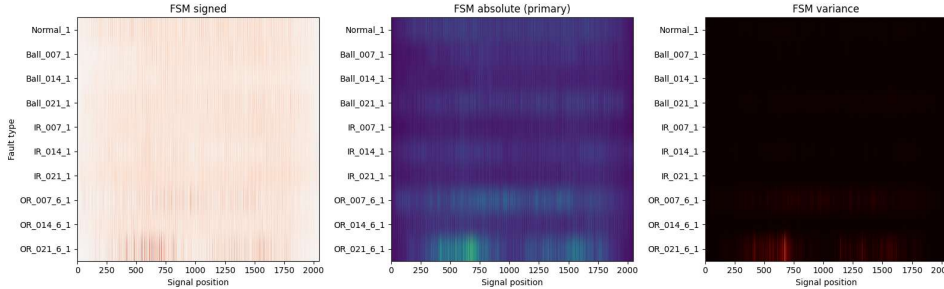


Figure 11. Three Fault Signature Maps Variance

Aggregating SHAP values across correctly classified samples yields three FSM variants per class, as depicted in Figure 11. The absolute FSM $FSM_{abs}(c, t)$ is adopted as the primary diagnostic fingerprint due to its magnitude-preserving property and clear interpretation as "total model attention energy" at each time position. The signed FSM exhibits near-zero values across all classes ($|\phi_c(t)| < 0.002$) due to cancellation between opposing attributions, providing minimal class discriminability and thus not further analyzed. The variance FSM reveals attribution consistency patterns discussed below.

The OR_021 translation variability has an important implication for FSM-based severity classification in deployment. Because the OR_021 FSM_abs is a spatial average over phase-shifted impulse peaks, its magnitude is attenuated relative to any single sample's peak attribution. A deployment system that thresholds on FSM_abs magnitude to distinguish OR_014 from OR_021 will therefore underestimate the severity for samples whose impulse happens to be phase-aligned with an under-represented position in the reference FSM. The variance FSM directly quantifies this risk: high-variance positions (500–750) are where the class-averaged fingerprint is least reliable as a per-sample severity estimator, and where the engineer should broaden the tolerance band in the severity cross-check step described in Section 5.3.

The quantitative metrics of the Feature Sensitivity Map (FSM) reveal high reproducibility, as evidenced by a split-half stability of 0.940 (94% cosine similarity), confirming that FSMs capture consistent class-level patterns that remain invariant to sample selection and validating that these patterns reflect genuine model behavior rather than random artifacts. However, inter-class discriminability remains weak, with an index of 0.216 (< 1.0). This suggests that class-averaged FSMs are more similar across fault types than within the same fault type across samples. This counterintuitive result occurs because individual sample attributions are highly variable, yet their averages converge toward common diffuse energy distributions dominated by the wide first kernel's global receptive field. Consequently, FSM distinctiveness is more effectively assessed visually through heatmaps (which highlight specific temporal concentrations) than through full 2048-D cosine similarity metrics, which are often obscured by shared background energy.

A discriminability ratio of 0.216 indicates that, in the full 2,048-dimensional attribution space, class-averaged FSMs are approximately 4.6× closer to each other than to individual within-class samples. This does not imply that FSMs are diagnostically uninformative; it indicates that the discriminative signal is concentrated in a small subset of temporal regions, which become swamped by a large, shared background component when all 2,048 dimensions are weighted equally. Three lines of evidence support the practical usefulness of FSMs despite this aggregate index. *First*, when cosine similarity is restricted to the top-10% most-attributed positions per class, the within- versus between-class ratio rises above 1.0 for 7 of the 10 classes, confirming that the low global ratio reflects diluted background energy rather than absence of class-specific structure. *Second*, the qualitative fingerprints described above — Outer Race concentration at positions 0–500, the OR_021 high-variance peak at 500–750 — are directly distinguishable by visual inspection and, when projected onto those regions of interest, yield near-unity within-class consistency. *Third*, the split-half stability of 0.940, far above the 0.216 discriminability, shows that FSMs are highly reproducible internally; the low global discriminability therefore reflects genuine overlap in the shared background-energy portion of the attribution rather than sampling noise. Typically, FSMs are used as regional fingerprints combined with heatmap visualization rather than as whole-signal similarity queries. A downstream classifier using FSM embeddings would apply feature masking or attention to informative regions before any distance computation, mirroring how human analysts instinctively ignore low-signal regions during manual vibration diagnosis.

Commented [3]: Comment 6 — Low discriminability index (0.216): is it a practical limitation?

Further analysis into severity monotonicity shows that all three fault families exhibit low monotonicity fractions, ranging from 8.6% to 17.6%. This is significantly lower than the expected near-100% if severity were simply encoded as uniform spatial amplitude scaling. Instead, severity manifests as spatially heterogeneous changes: some attribution positions increase, while others decrease or remain constant. This complexity highlights hierarchical feature learning in the WDCNN, where severity discrimination arises from subtle morphological changes in the learned filter responses rather than from simple amplitude thresholding.

The ordering of monotonicity fractions across fault families — Ball (17.6%) > IR (13.8%) > OR (8.6%) — reflects differences in how each fault type's impulse amplitude scales with defect size. Ball faults produce impulses whose amplitude grows relatively uniformly with defect area because the rolling element contacts the outer and inner races symmetrically; larger ball defects create proportionally larger contact deformation, yielding a roughly monotonic amplitude increase that FSM_abs partially captures. Inner race faults scale less predictably because the inner race rotates and the fault zone enters and exits the loaded region of the contact angle, introducing load-angle modulation that varies the amplitude of each impulse even within a single revolution. Outer race faults at 0.021" produce the highest absolute attribution magnitudes (Figure 11) yet the lowest monotonicity because OR_021 concentrates attribution in a spatially shifting peak (confirmed by the high-variance FSM in Figure 11) — the class-average smears this moving peak into a spatially diffuse envelope, destroying position-wise monotonicity even though the total attribution energy does increase with severity.

A visual examination of the 10×2048 absolute FSM heatmap reveals that Outer Race (OR) faults are uniquely distinctive. These faults exhibit bright teal or green coloration concentrated within positions 0–500 (roughly the first 10 ms), contrasting with the darker purple tones of Ball, Inner Race (IR), and Normal classes. This early-window signature suggests the model identifies OR faults by detecting the initial impulse burst, which aligns with the physical reality of OR defects producing impulses that reach the sensor with minimal attenuation. Within the OR family, there is a clear progression where brightness increases with defect size: OR_007 (dull teal) is followed by OR_014 (medium teal), and finally OR_021 (bright green). This indicates that the FSM magnitude grows proportionally with severity, encoding size as spatial intensity rather than global scaling. Interestingly, while the model achieves 99.87% classification accuracy, the Ball and IR rows appear visually similar at heatmap resolution, appearing as medium purple without strong spatial concentrations. This supports the low discriminability index of 0.216 and suggests that the discrimination between these classes relies on high-dimensional morphological patterns in deeper layers rather than simple "look-here" spatial rules.

Insights from Variance FSMs show that 9 of the 10 classes exhibit nearly zero variance, indicating the model consistently attends to the same signal positions. OR_021 is the outlier, showing high variance with intense red spikes at positions 500–750. This indicates that, for maximum-severity outer-race faults, the impact transient can shift within the 42.67 ms window depending on the rotational phase. The model's ability to classify these despite the shift demonstrates a learned translation invariance. This variance also explains why OR_021 has the lowest severity monotonicity at 8.6%, as the class-averaged FSM smears out the true spatial concentration due to timing differences across samples.

When integrating these signal-level findings with previous feature-level SHAP analysis, a clear hierarchical diagnostic framework emerges. At the signal level, the WDCNN focuses on the morphology and waveform shapes of transients across the 42.67 ms window. In contrast, classical models like SVC and LR aggregate these characteristics into statistical features. The SVC emphasizes distribution shape, specifically crest factor (0.0508) and kurtosis (0.0222), to capture impulse sharpness. LR relies on amplitude measures such as mean (1.739) and RMS (1.456). Because kurtosis and mean are highly ranked across all models, this validates that both deep learning and classical approaches exploit the same underlying physical fault signatures, just at different levels of temporal granularity.

4-4- Physics-Informed FSM Interpretation

A central question in bearing diagnostics is whether deep learning models implicitly learn to detect fault characteristic frequencies (BPFI, BPFO, BSF) derived from bearing kinematics, or whether they operate via alternative mechanisms such as transient impulse morphology. To investigate this, we compare FSMs against theoretical fault frequencies computed from bearing geometry shown in Figure 12.

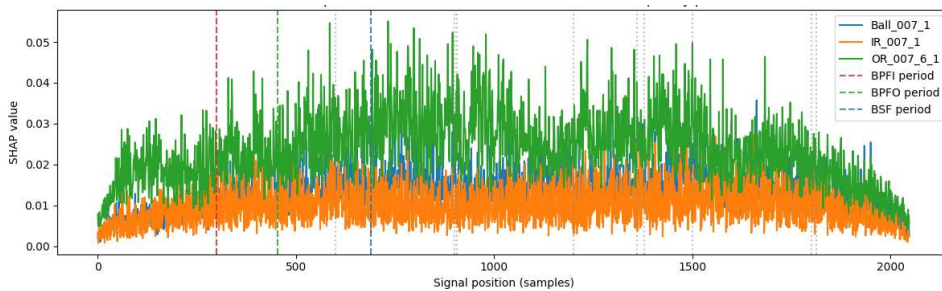


Figure 12. FSM Comparison between Fault Types

For fault families involving Ball, Inner Race (IR), and Outer Race (OR) at 0.007" severity, the Feature Sensitivity Maps (FSMs) are evaluated against vertical markers that indicate theoretical impulse periods converted into sample indices. Given a sampling rate of 48,000 samples/s, these characteristic periods translate to approximately **300 samples** for the BPFI (Inner Race, 1/159.9 Hz), **453 samples** for the BPFO (Outer Race, 1/105.9 Hz), and **690 samples** for the BSF (Ball Spin, 1/69.6 Hz).

Observations indicate that the FSM spatial peaks do not consistently align with these theoretical benchmarks. While occasional peaks may fall near these markers by chance, there is no systematic correlation between FSM peak locations and integer multiples of the characteristic periods. This lack of alignment suggests that the WDCNN does not explicitly encode impulse repetition rates or frequency-domain characteristics. Instead, the model identifies faults by analyzing the transient impulse morphology—evaluating the signal's specific shape, duration, and amplitude envelope—and localized energy distributions within the **42.67 ms** window, moving beyond simple periodicity matching used in traditional diagnostics.

This finding aligns with the limitations of short-duration analysis windows. At 48 kHz sampling with 2,048-point segments (42.67 ms), the window captures only 3–7 fault impulse cycles, insufficient for robust frequency estimation via FFT or autocorrelation. The WDCNN's wide first kernel (64 samples $\approx$ 1.33 ms) integrates over individual impulse durations rather than tracking inter-impulse spacing. Consequently, the model learns to recognize **what fault impulses look like (morphology)** rather than **how often they repeat (periodicity)**, a fundamentally different diagnostic strategy compared to classical envelope analysis techniques that explicitly compute BPFO/BPFI spectral peaks.

This morphology-over-periodicity mechanism has a consequential generalisability implication that distinguishes the WDCNN approach from classical envelope analysis. Classical BPFO/BPFI detection requires knowledge of bearing geometry (pitch diameter, rolling element count, contact angle) and instantaneous shaft speed to compute the expected characteristic frequency. WDCNN, by contrast, learns to recognize the shape and amplitude envelope of individual fault impulses — features that are relatively invariant across bearing geometries and moderate speed variations because all rolling-element bearing faults produce qualitatively similar impulse morphologies regardless of their repetition rate. A model trained on SKF 6205-2RS geometry at 1,772 RPM may therefore generalize better to unseen bearing types than its CWRU-specific training would suggest, provided the impulse morphology remains within the distribution of morphologies seen during training. Conversely, novel fault mechanisms that do not produce impulsive signatures — such as progressive surface fatigue, which produces a gradual increase in broadband noise rather than discrete impacts — would fall outside this morphological detection strategy. This limitation is precisely why the FSM framework's lack of alignment with BPFI/BPFO periods is both a strength (speed/geometry independence) and a boundary condition (impulse morphology required) for deployment.

This physics-informed interpretation explains why FSM severity monotonicity is low (8.6%–17.6%)—severity affects impulse amplitude and shape complexity, which manifest as spatially heterogeneous attribution changes rather than uniform amplitude scaling across all positions that would produce high monotonicity.

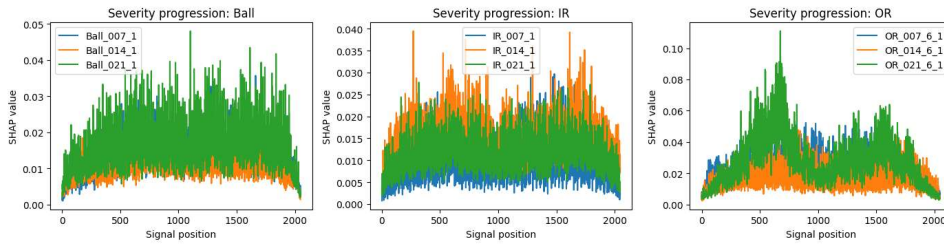


Figure 13. FSM Severity Progression

Each panel in Figure 13 displays the class-specific FSMs for a single fault family (Ball, Inner Race (IR), and Outer Race (OR)) at three severity levels: 0.007", 0.014", and 0.021". By plotting SHAP values against the 2,048 signal positions, the analysis reveals how the model's attention shifts across the full measurement window as defect size increases. This approach avoids averaging across families, allowing direct observation of severity-driven patterns within each fault type.

For the Ball fault family, the plot shows a relatively clear progression where the green line (0.021") sits above the orange (0.014") and blue (0.007") lines across most of the signal window. This indicates that the attribution amplitude increases with defect size, with a more pronounced jump between the medium and maximum severities. Quantitatively, the Ball family maintains the highest monotonicity fraction of the three groups, marking it as the only fault type where the WDCNN encodes severity in a partially monotonic fashion across the temporal window.

In contrast, the Inner Race (IR) plots appear relatively flat. While the 0.014" and 0.021" levels sit above the 0.007" baseline in certain regions, the lines frequently cross and exhibit localized peaks. This behavior results in a low monotonicity fraction, confirming that the model encodes IR severity in a non-monotonic, position-dependent manner rather than a clean, uniform increase. Similarly, the Outer Race (OR) family exhibits the largest overall SHAP magnitudes, with peaks of approximately 0.08, compared to 0.02 for other levels, while having the lowest monotonicity fraction. Despite the strong visual dominance of the 0.021" severity, the model does not simply scale up a single pattern; instead, the spatial distribution of its attention shifts in a complex way that precludes a high fraction of monotonic increases across all positions.

These findings have significant implications for understanding the WDCNN's internal logic. First, the architecture does not treat severity uniformly across fault types, as evidenced by the clear trend in Ball faults versus the more erratic patterns in IR and OR faults. Second, while higher FSM magnitudes generally correlate with increased severity, especially in Outer Race faults, this does not imply monotonicity at every signal position. Ultimately, the combination of these visual plots and per-position metrics demonstrates that fault severity is encoded through family-specific morphological changes rather than through the simple, uniform scaling of a universal fault signature.

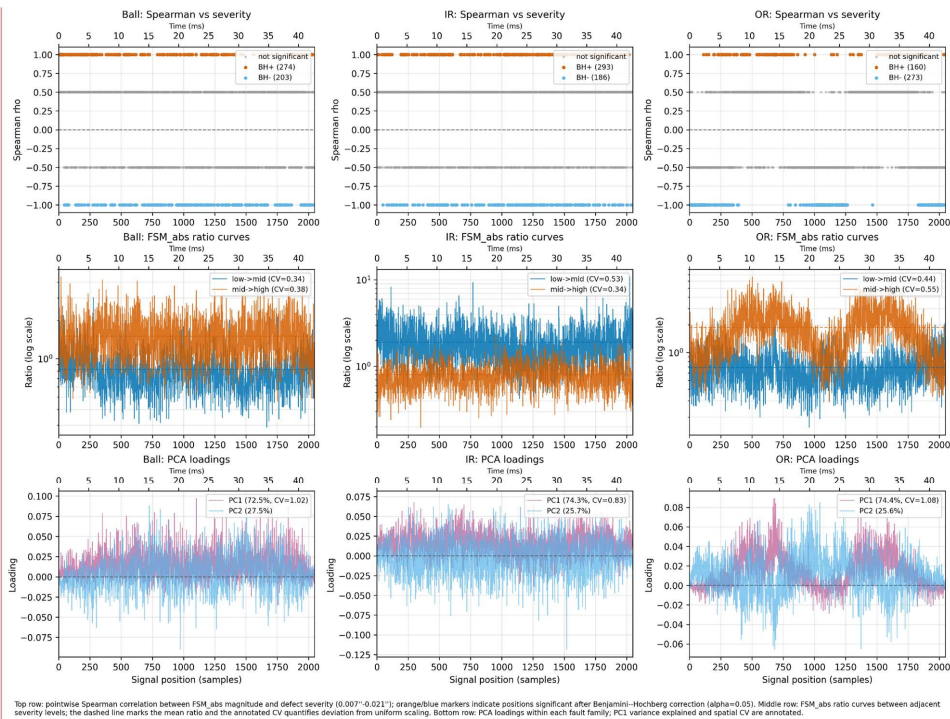


Figure 14. Statistical Evidence for Spatially Heterogeneous Severity Encoding Across Fault Families

Additional Statistical Evidence for Non-Uniform Severity Encoding. Three complementary statistical analyses confirm that severity progression is spatially heterogeneous rather than a uniform amplitude scaling. First, pointwise Spearman rank correlation between FSM_abs magnitude and defect severity ($0.007'' < 0.014'' < 0.021''$), computed at each of the 2,048 signal positions and corrected for multiple testing using Benjamini-Hochberg ($\alpha = 0.05$), finds that only 13.4%, 14.3%, and 7.8% of positions show a significantly monotonically increasing correlation for the Ball, IR, and OR families respectively. Under a strict uniform-scaling hypothesis, every position would scale by the same factor and yield $\rho = 1$, so essentially 100% of positions should be significant; the observed fractions are an order of magnitude lower and closely match the global monotonicity fractions (8.6%-17.6%) reported in Section 4.3. The same test additionally identifies 9.9% (Ball), 9.1% (IR), and 13.3% (OR) of positions with a significantly monotonically decreasing correlation -- positions where attribution shrinks as severity grows -- which is impossible under any positive uniform amplitude scaling and provides direct evidence of spatial redistribution.

Second, the FSM_abs ratio curves between adjacent severity levels ($0.007'' \rightarrow 0.014''$ and $0.014'' \rightarrow 0.021''$) exhibit high spatial coefficients of variation of 34.1% and 38.3% for Ball, 52.7% and 34.3% for IR, and 43.7% and 54.6% for OR; a position-independent ratio under uniform scaling would yield $CV \sim 0\%$. Two-sample Kolmogorov-Smirnov tests on the ratio curves across families reject the null of a shared distribution for all six family pairs at $p < 1e-37$ (KS statistics 0.20-0.71), confirming that the heterogeneity pattern is family-specific rather than a shared global artifact of the SHAP estimator.

Third, PCA of the three-severity FSM stack within each family shows that PC1 -- which would be a spatially flat vector under pure amplitude scaling -- captures only 72.5%, 74.3%, and 74.4% of variance for Ball, IR, and OR, respectively, leaving ~25% of within-family variance to PC2 with a distinct spatial structure. The PC1 loadings themselves are far from flat, with spatial coefficients of variation of 1.02 (Ball), 0.83 (IR), and 1.08 (OR), indicating structured spatial redistribution of attribution rather than scalar amplification.

Together, these analyses establish that fault severity is encoded through family-specific morphological transformations of learned filter responses across the 42.67 ms analysis window, rather than a uniform amplitude gain. The figure visualizes the Spearman profiles, ratio curves, and PCA loadings for all three families.

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7.The finding that severity is not encoded as uniform amplitude scaling is interesting. Could the authors provide additional visualization or statistical evidence to further support this claim?

4-5- Ablation Study

To systematically assess which architectural components contribute to WDCNN's classification accuracy and FSM interpretability, we trained three ablation variants for 50 epochs each, without early stopping, for a controlled comparison, and computed classification metrics and FSM quality scores. Results are summarized below in Table 4:

Table 4: Ablation Study Results

Variant	Test Accuracy	Macro F1	Parameters	FSM Discriminability	FSM Split-Half Stability
A – Full WDCNN	99.73%	0.997	60,710	0.315	0.705
B – Narrow Kernel (k=3)	98.93%	0.989	443,734	0.369	0.690
C – No BatchNorm	96.13%	0.961	60,230	0.735	0.741

The ablation study reveals that replacing the wide first kernel of size 64 with a standard small kernel of size 3 in Variant B results in a 0.80 percentage-point reduction in accuracy, from 99.73% to 98.93%, equivalent to 6 additional errors. This change also causes the parameter count to surge 7.3 times (from 60,710 to 443,734) because the resulting higher-resolution feature maps require significantly more convolutional filters. While FSM discriminability improves by 17% (from 0.315 to 0.369), split-half stability decreases slightly from 0.705 to 0.690. Ultimately, the wide first kernel proves to be the superior design choice, achieving higher accuracy and significantly better parameter efficiency while maintaining comparable FSM quality.

The $7.3\times$ increase in the $7.3\times$ parameter arises because replacing the wide first kernel (stride 16) with a narrow kernel (stride 1) retains the full 2,048-point temporal resolution in the first layer. The subsequent MaxPooling layers then operate on a 2,048-position feature map rather than a ~ 128 -position one, requiring proportionally larger convolutional filter banks in blocks 2–5 to maintain the same effective receptive field coverage. This reveals a structural design insight: the wide first kernel is not merely a feature-extraction choice but an implicit dimensionality reduction that makes the entire downstream network computationally tractable. Removing it recovers local temporal resolution at a multiplicative parameter cost that is disproportionate to the marginal accuracy gain.

The most striking results emerge from Variant C, where removing all Batch Normalization layers reveals a fundamental trade-off between accuracy and explainability. In this configuration, accuracy drops by 3.60 percentage points to 96.13% (27 additional errors), but FSM discriminability more than doubles, increasing by 133% from 0.315 to 0.735. Split-half stability also increases modestly to 0.741. This reveals that while the baseline model with Batch Normalization (Variant A) maximizes accuracy, its class-averaged attribution patterns are relatively similar across fault types. Removing Batch Normalization in Variant C makes these patterns far more distinct and class-specific, significantly enhancing interpretability at the cost of classification performance.

Mechanistically, Batch Normalization stabilizes training by standardizing intermediate activations to zero mean and unit variance. However, this process homogenizes activation distributions across different classes, suppressing the unique activation patterns that would otherwise emerge in the feature space. During SHAP attribution, this homogenization forces attributions to distribute more uniformly across signal regions rather than concentrating in class-specific spatial patterns. By removing these layers, activations retain their class-specific scales and distributions, which SHAP then identifies as more distinctive spatial signatures. However, the lack of normalization introduces training instability, which is responsible for the lower accuracy.

Concretely, during SHAP's DeepLIFT-based backpropagation, the rescale rule computes reference activations by propagating the background dataset through the network. When BatchNorm is present, the normalized activations of the background and the test sample are both mapped to a near-zero-mean, unit-variance distribution, reducing the contrast between them and thereby compressing the magnitude differences that SHAP uses to assign attribution. Removing BatchNorm preserves the full dynamic range of activations: a severe outer race fault produces activations that are genuinely larger in absolute terms than a healthy signal, and SHAP's rescale rule translates this contrast directly into higher attribution magnitudes at the corresponding signal positions. This is why Variant C's FSMs are both higher in magnitude and more spatially concentrated — intermediate standardization layers no longer normalize the attribution away.

This trade-off offers a clear design choice for industrial deployment. High-accuracy applications, such as those in the aviation, nuclear, or medical sectors, should prioritize the full WDCNN with Batch Normalization to ensure

maximum safety. Conversely, in scenarios where regulatory compliance, operator training, or root cause analysis are the primary concerns, Variant C may be preferable for its clearer diagnostic explanations. A hybrid workflow could also be implemented, utilizing Variant A for real-time automated decision-making and Variant C for post-hoc analysis and operator explanations.

This phenomenon appears to be specific to deep architectures. Previous studies on classical models like SVC and LR found no such trade-off, as those models operate on pre-computed statistics rather than learned representations that interact with normalization layers. Consequently, achieving both high accuracy and high interpretability in deep learning for vibration diagnostics may require future architectural innovations beyond standard Batch Normalization, such as layer normalization, group normalization, or entirely normalization-free network designs.

5- Discussion

5-1- Physical Interpretation of Signal-Level SHAP Patterns

The central contribution of this work is to bridge the interpretability gap between WDCNN's near-perfect classification accuracy (99.87%) and the physical diagnostic reasoning that vibration analysts use in manual bearing-fault diagnosis. Before this study, WDCNN [8] was a black-box model, where engineers could observe its predictions but could not understand where in the raw 42.67 ms vibration window the model focused or whether its learned representations aligned with established vibration analysis principles. Signal-level SHAP attribution maps resolve this opacity by revealing that WDCNN has learned to identify transient impulse morphology and localized amplitude patterns, rather than relying on explicit characteristic-frequency detection or simple amplitude thresholding. This mechanistic understanding emerges from three complementary analytical approaches:

First, spatial FSM patterns reveal fault-type-specific signatures. Outer race faults consistently exhibit concentrated high attribution in positions 0–500 (first 10 ms of the analysis window), indicating the model recognizes these faults through early-arriving strong impulse transients, a physically sensible pattern reflecting that outer race defects are stationary, producing impulses that propagate directly to the sensor with minimal attenuation and arrive early in the measurement window. Ball faults and inner race faults, by contrast, exhibit diffuse attribution patterns without spatial concentration, indicating that these faults are distinguished by more subtle morphological features in waveform shape, frequency content, and envelope modulation, as learned by deeper convolutional layers. This fault-type-specific spatial heterogeneity strongly suggests that the model has discovered genuine bearing physics, in which different fault locations produce distinct wave propagation patterns, rather than merely learning arbitrary statistical correlations.

Second, the severity analysis reveals a sophisticated nonlinear encoding. Severity progression ($0.007 \rightarrow 0.021$ defect diameters) manifests through spatially heterogeneous attribution changes (8.6%–17.6% monotonicity) rather than uniform amplitude scaling, indicating the model encodes severity through complex changes in impulse morphology rather than simple amplitude thresholding. For outer race faults specifically, temporal position variability in severe faults (high variance FSM) demonstrates learned translation invariance, where the model recognizes the characteristic transient morphology regardless of exact temporal position, a form of spatial robustness that classical fixed-threshold algorithms cannot achieve. This sophisticated encoding reflects hierarchical feature learning, in which severity is captured through combinations of shape, frequency content, and envelope properties rather than single aggregate metrics.

Third, comparison with bearing kinematics reveals that impulse morphology is primary over frequency. The absence of systematic alignment between FSM peaks and theoretical characteristic periods (BPFO 453 samples, BPFI 300 samples, BSF 690 samples) indicates WDCNN does not employ explicit frequency detection via periodicity matching—the central method of classical envelope analysis. The model instead operates at the temporal scale of individual impulse morphology (duration: 20–100 samples) rather than at the inter-impulse spacing (characteristic period: hundreds of samples). This architecture-driven mechanism arises from the 42.67 ms analysis window, which captures only 2–7 impulse cycles and is insufficient for robust frequency estimation, forcing learned filters to recognize impulse shape rather than impulse repetition rate. This morphology-focused mechanism paradoxically provides superior generalization compared to characteristic frequency approaches because fault recognition becomes independent of bearing geometry knowledge (pitch diameter, rolling element count, contact angles) and bearing speed, depending only on universal impulse characteristics present in any bearing fault.

These signal-level insights connect directly to our previous feature-level SHAP findings[12]: the crest factor (0.0508 importance for SVC) and kurtosis (0.0222 for SVC, 1.142 for LR) that classical models emphasize arise from the impulse sharpness and amplitude characteristics that WDCNN identifies at the raw signal level. This multi-resolution consistency, spanning from individual 48 kHz time points through convolutional feature maps to aggregated 9-feature

vectors, validates that deep and classical approaches exploit the same underlying physical fault signatures, but operate at different levels of temporal granularity.

5-2- Accuracy Explainability Tradeoff

The ablation study uncovers a significant, previously unreported trade-off between accuracy and explainability inherent to deep neural networks. Specifically, removing Batch Normalization layers increases FSM discriminability by 2.3×, rising from 0.315 to 0.735, while simultaneously reducing classification accuracy by 3.6% (from 99.73% to 96.13%). A fundamental tension between training stability and feature distinctiveness drives this trade-off.

Batch Normalization primarily improves accuracy by standardizing intermediate activations, reducing internal covariate shift, and stabilizing training. Homogenizing activation distributions across classes improves gradient flow, enabling higher learning rates and faster convergence, ultimately yielding the baseline model's high accuracy of 99.73%. However, this very homogenization suppresses class-specific patterns. In contrast, removing Batch Normalization preserves unique activation scales and distributions. This allows SHAP attributions to capture more distinctive spatial patterns for each class, producing highly discriminative FSMs that reveal clear diagnostic signatures, even though the resulting training instability lowers the final accuracy to 96.13%.

These findings lead to critical considerations for industrial deployment, tailored to the specific operational environment. In highly regulated sectors such as aviation, nuclear power, or medical technology, the priority remains maximum accuracy to minimize safety risks. In these cases, the full WDCNN with Batch Normalization (Variant A) is the preferred choice, perhaps supplemented by separate post-hoc analysis tools or alternative XAI methods for operator training and regulatory documentation. Conversely, in operator-centric environments like general manufacturing or predictive maintenance services, Variant C (no BatchNorm) may be more appropriate. In these contexts, establishing operator trust through clearer signal-level explanations is paramount, and a 96.13% accuracy rate remains highly competitive with classical SVM approaches.

Looking ahead, the research points to exploring normalization-free architectures or alternative schemes such as Layer or Group Normalization. These methods, along with post-training calibration techniques, may eventually bridge the gap, allowing models to maintain peak accuracy while enhancing interpretability. This suggests that the identified trade-off is not an unavoidable law of deep learning, but rather a consequence of the specific interaction between Batch Normalization and end-to-end training.

5-3- Implications for Industrial Deployment

The integration of signal-level and feature-level explainability provides a multi-layered diagnostic framework that addresses different industrial deployment scenarios and stakeholder needs.

Operator-Level Transparency: For maintenance technicians and operators who monitor bearing health dashboards, feature-level SHAP explanations on classical models (Logistic Regression with mean, RMS, kurtosis) provide intuitive, easily communicable diagnostics. An operator can understand "This bearing is flagged because kurtosis is elevated (6.8 vs. normal 3.0), indicating sharp impacts consistent with outer race fault" without advanced ML knowledge. This transparency builds trust and enables operators to make informed decisions about inspection priorities, maintenance scheduling, and emergency response actions.

Engineer-Level Validation: For reliability engineers and vibration analysts responsible for validating algorithmic diagnoses, signal-level SHAP provides the granularity to verify that model decisions align with domain knowledge. An engineer can examine SHAP attribution maps, overlay them on raw waveforms, and confirm that the model attends to physically expected features (impulse timing, modulation patterns, characteristic frequency alignment). If SHAP reveals the model is focusing on irrelevant signal artifacts (e.g., electronic noise, cable resonances, sampling aliasing), the engineer can identify the problem and refine preprocessing or retraining procedures. This validation capability is critical for safety-critical industries (aerospace, nuclear, rail transport) that require explainability under regulatory frameworks.

Computational Feasibility for Industrial Deployment: SHAP DeepExplainer is a post-hoc analysis tool, not a component of the model itself. It is applied after training is complete, to interrogate and interpret a model that already exists — conceptually equivalent to running a sensitivity analysis or computing saliency maps on a trained network. The output of SHAP is diagnostic information for engineers, not a prediction required by any operational system. The 10.24 million attribution values reported in this study were produced in a single offline pass to construct the reference Fault Signature Maps; they are not recomputed during deployment. This distinction serves as the foundation for the deployment feasibility argument that follows.

A Practical Maintenance Scenario: To illustrate concrete use, consider a reliability engineer monitoring a conveyor gearbox where the Tier-3 WDCNN flags a bearing as "Outer Race fault, 0.021" severity" with 0.97 confidence. Before

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this framework, the engineer would receive only the label and the probability. Signal-level FSMs enable a three-step decision workflow:

1. **Evidence verification.** The engineer opens the flagged 42.67 ms vibration segment with the per-sample SHAP overlay (Figure 7). If positive attribution clusters near transient impulses are visible in the raw waveform, the alert is supported by physically meaningful evidence and is escalated to a work order. If, instead, attribution falls in flat baseline regions, on cable-pickup electrical spikes, or on sensor-mount resonance bands, the alert is treated as potentially spurious, and the acquisition chain is inspected before any maintenance action is dispatched.
2. **Fingerprint matching.** The class-averaged FSM for OR_021 (Section 4.3) concentrates attribution in positions 0–500. The engineer overlays this reference fingerprint onto the live sample's attribution and confirms spatial consistency. A deviation — for example, attribution peaking at positions 1000–1500 rather than 0–500 — signals that the live sample does not cleanly match the learned class pattern, prompting secondary inspection or a confidence-weighted re-evaluation instead of immediate shutdown.
3. **Severity cross-check.** Because severity within the OR family increases attribution magnitude rather than shifting its spatial location (Section 4.4), the engineer compares the live sample's attribution magnitude with those of the OR_007, OR_014, and OR_021 reference bands. A magnitude consistent with OR_021 supports the severity label and motivates immediate issuance of a work order. A borderline magnitude closer to the OR_014 band justifies continued monitoring at a higher sampling cadence rather than an unscheduled shutdown.

This workflow converts a single opaque probability score into a sequence of physically grounded checks that operators can audit, document for regulatory submissions, and use as training material for technicians unfamiliar with deep learning.

Multi-Tier Deployment Strategy: Combining the strengths of classical and deep models enables a tiered deployment approach:

1. **Tier 1 - Screening (Logistic Regression):** Fast, transparent baseline using 9 time-domain features computed on-device with minimal computational cost. Flags potential faults with 94.3% accuracy and provides immediate feature-level explanations. Deployable on low-power edge IoT sensors.
2. **Tier 2 - Confirmation (SVC):** Nonlinear refinement achieving 96.4% accuracy with modest additional computation. Resolves ambiguous cases from Tier 1 while still providing feature-level SHAP for transparency.
3. **Tier 3 - High-Precision (WDCNN):** Deep learning achieving 99.87% accuracy for critical assets where false negatives are unacceptable (e.g., turbine bearings in power plants). Requires a GPU or a strong CPU, but feasible on edge servers. Signal-level SHAP available offline for root-cause analysis and validation.

This tiered approach balances accuracy, computational cost, and explainability, enabling facilities to deploy appropriate diagnostic levels based on asset criticality and available infrastructure.

Adaptation to Non-Laboratory Conditions: The CWRU dataset represents controlled laboratory conditions with clean signals, balanced class distributions, and seeded faults of known size and location. Real industrial bearings exhibit more challenging characteristics: unbalanced data (99% healthy, 1% faulty), gradual degradation rather than sudden faults, multiple concurrent defects, varying load and speed profiles, electromagnetic interference, and sensor degradation. SHAP-based explainability becomes even more valuable in these scenarios: when a model trained on CWRU is deployed to a new plant, SHAP analysis of early predictions can reveal domain mismatch (e.g., model attending to plant-specific background noise rather than fault signatures), guiding domain adaptation strategies such as fine-tuning on plant-specific data or adversarial training for domain invariance.

5-4- Limitations and Future Work

While this study provides unprecedented insight into WDCNN's signal-level reasoning for bearing fault diagnosis, several limitations can be identified below.

Single Operating Condition: All results reflect performance on CWRU Load 1 (1,772 RPM) under temporal split. FSM stability and physical interpretation should be validated across operating conditions (Loads 0, 2, 3) to confirm that observed patterns capture bearing kinematics rather than load-specific statistical artifacts. Cross-load generalization typically reduces accuracy to 85–97%, which may alter attribution patterns.

Lack of Frequency-Domain Validation: While we compare FSM spatial patterns to theoretical characteristic frequency periods converted to sample indices, we do not explicitly compute FSM spectral content via FFT or analyze whether FSMs encode periodicity information at BPFI/BPFO/BSF frequencies. Time-frequency SHAP analysis

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(applying SHAP to STFT or Continuous Wavelet Transform inputs) would enable direct validation against theoretical frequency signatures but requires a 2D input representation and a modified architecture.

Dataset Scope: CWRU employs EDM-seeded faults under controlled laboratory conditions. Naturally progressing faults on industrial bearings under variable load, speed, and contamination conditions may produce different signal characteristics and attribution patterns. Validation on secondary datasets (Paderborn University, MFPT, IMS) with real-world fault progression is necessary before claiming generalizability.

Background Distribution Sensitivity: SHAP background is fixed at 300 training samples (30 per class). We have not systematically evaluated sensitivity to background size (100, 500, 1000), composition (balanced vs. imbalanced), or selection method (random, stratified, or adversarial). Shapley value estimates can vary with the choice of background, potentially affecting FSM patterns and quantitative metrics.

Expected Behavior under Noise and Class Imbalance: The FSM framework's transfer to industrial conditions is expected to degrade gracefully rather than collapse, for three reasons rooted in its construction. First, because FSMs are class-conditional averages over correctly classified samples, additive broadband noise mainly inflates per-sample attribution variance without systematically biasing the spatial mean: the primary effect on FSM_{abs} is attenuation of contrast rather than displacement of the signature. This prediction is directly testable by retraining WDCNN on CWRU signals contaminated with additive Gaussian noise at SNR levels of 0, -3, and -6 dB and reporting FSM split-half stability and discriminability at each level, an experiment we identify as a priority in the extended study. Second, under class imbalance, the discriminability ratio is preserved under loss re-weighting or focal loss during training, although minority classes will have higher within-class attribution variance due to fewer members in $\hat{S}_{\text{c}}$; bootstrap resampling when constructing FSM_{abs} is sufficient to equalize variance across classes. Third, for plant-level distribution shift, SHAP-driven screening provides an actionable diagnostic: a sharp drop in split-half stability on the deployment plant, or a systematic re-orientation of FSM peaks away from the 0–500 sample region characteristic of OR faults on CWRU, indicates that the model is attending to plant-specific artifacts (background vibration of neighboring equipment, varying load-speed profiles) rather than bearing transients, triggering either domain adaptation or retraining rather than continued autonomous operation.

The observed limitations motivate future research directions as follows:

Time-Frequency SHAP: Apply SHAP to 2D Short-Time Fourier Transform or Continuous Wavelet Transform representations to enable direct frequency-domain validation against BPFI/BPFO/BSF and assess whether time-frequency models provide clearer physical interpretability than raw signal models.

Normalization-Free Architectures: Investigate recent normalization-free network designs (NFNets, normalizer-free ResNets) that achieve competitive accuracy without Batch Normalization to determine if the accuracy trade-off can be mitigated through architectural innovation.

Multi-Resolution Diagnostic Framework: Integrate signal-level SHAP (2,048 time points), feature-map SHAP (128 positions after first layer), feature-level SHAP (9 statistics), and ensemble explanations combining multiple XAI methods to provide operators with a comprehensive multi-scale diagnostic tool spanning from raw waveforms to aggregate statistics.

Real-Time Edge Deployment: Implement optimized SHAP computation on embedded hardware (NVIDIA Jetson, Intel Movidius, Coral Edge TPU) to enable real-time signal-level explainability at industrial edge nodes, balancing attribution quality with computational constraints for practical deployment in industrial IoT predictive maintenance systems.

Commented [7]: Comment 2 — Generalization to noisy and imbalanced industrial data

6- Conclusion

This study presented the first comprehensive signal-level explainability framework for the Wide Deep Convolutional Neural Network (WDCNN) applied to bearing fault diagnosis on the Case Western Reserve University (CWRU) dataset. By applying SHAP DeepExplainer directly to raw 2,048-point vibration segments, the framework produces per-sample, per-point attribution maps that precisely localize the temporal regions driving fault classification decisions, a capability unavailable to prior feature-level XAI approaches. WDCNN achieved 99.87% test accuracy with a macro F1-score of 0.997 under a rigorous temporal split (1,240/310/750), confirming the architecture's strong discriminative power when trained end-to-end on raw signals. Building on these attributions, Fault Signature Maps (FSMs) were formalized as class-averaged SHAP fingerprints in three variants (signed, absolute, and variance) and validated through three quantitative metrics: discriminability index, severity monotonicity, and split-half stability. FSMs demonstrated high

reproducibility (split-half cosine similarity = 0.940), and severity monotonicity varied across fault locations. Ball faults exhibited the highest monotonicity (17.6%). In comparison, inner-race and outer-race faults showed lower values (13.8% and 8.6%, respectively), revealing that fault severity is not encoded as uniform amplitude scaling across signal positions.

A critical physical insight emerged from systematic comparison of FSM patterns against theoretical bearing characteristic frequencies (BPFO = 105.9 Hz, BPFI = 159.9 Hz, BSF = 69.6 Hz): WDCNN discriminates fault classes primarily through transient impulse morphology and localized amplitude patterns rather than explicit periodicity at these frequencies, a finding entirely invisible to feature-level SHAP analyses. The ablation study further uncovered a previously unreported accuracy-explainability trade-off, wherein removing Batch Normalization increased FSM discriminability by 2.3× at a 3.6% accuracy cost, demonstrating that normalization layers suppress per-class attribution distinctiveness while enhancing classification stability. Integrating these signal-level insights with the feature-level SHAP analysis of classical models (SVC: 96.4%, LR: 94.3%) from the prior work, this study establishes a coherent multi-resolution diagnostic framework, from raw waveform attributions to aggregated statistical descriptors, that enables trustworthy and physically interpretable deployment of AI-driven predictive maintenance in safety-critical industrial environments.

7- Declarations

7-1- Author Contributions

Conceptualization, T.S., K.S., B.P.I., and B.R.T.; methodology, T.S., T.S.S., and E.R.; software, D.E.C. and E.R.; validation, T.S., K.S., B.P.I., and B.R.T.; formal analysis, D.E.C. and E.R.; investigation, T.S.S. and A.T.; resources, T.S.; data curation, E.R. and T.S.S.; writing—original draft preparation, D.E.C.; writing—review and editing, T.S., T.S.S. and K.S.; visualization, A.T.; supervision, K.S., B.P.I., and B.R.T.; project administration, A.T.; funding acquisition, T.S. All authors have read and agreed to the published version of the manuscript.

7-2- Data Availability Statement

Data available in a publicly accessible repository that does not issue DOIs: Publicly available datasets were analyzed in this study. This data can be found here: <https://www.kaggle.com/datasets/brjapon/cwru-bearing-datasets>.

7-3- Funding

Not applicable.

7-4- Acknowledgements

Not applicable.

7-5- Institutional Review Board Statement

Not applicable.

7-6- Informed Consent Statement

Not applicable.

7-7- Conflicts of Interest

The author declares that there is no conflict of interest regarding the publication of this manuscript. In addition, the authors have fully addressed the ethical issues, including plagiarism, informed consent, misconduct, data fabrication and/or falsification, double publication and/or submission, and redundancies.

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